

Research paper

A printed circuit board surface defect detection method for long-tail and multi-scale scenarios

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ABSTRACT

Surface defects on Printed Circuit Board (PCB) in industrial production exhibit characteristics of random occurrence, uneven category distribution, variable scales, and minute dimensions, increasing the difficulty of quality inspection. To achieve multi-scale defect detection on PCB surfaces in long-tail small-target scenarios, the Long-Tail Dynamic Multi-Scale Printed Circuit Board (LDM-PCB) detection approach proposed in this paper employs the Long-Tail Feature Extraction Network (LTFE-Net) as the backbone, enhances the representation of tail defects by the Adaptive Tail Attention (ATA) module, improves the ability of the model to quickly capture the low-frequency defect features, and effectively solves the imbalance problem of feature learning under long-tail data distribution. The Dynamic Multi-Scale Fusion (DMS-Fuse) architecture dynamically adjusts feature fusion weights for defects of varying sizes through adaptive weighting strategies, enabling feature interaction across scales. A designed dynamic prediction layer preserves high-resolution defect features, directly outputting dynamic information to mitigate detail degradation in deep networks and improve localization accuracy for subtle defects. On self-built long-tail defect dataset, LDM-PCB achieves 99.1% mean Average Precision at Intersection-over-Union threshold 0.5 ($mAP_{0.5}$) with only 8.61 million (M) parameters, surpassing baseline models by 1.8 percentage points. The detection speed reaches 100 frames per second (FPS), achieving a balance between accuracy and speed, with results superior to other algorithms. Generalization experiments on public PCB datasets further demonstrate the optimal performance of LDM-PCB. Deployment results on edge devices indicate industrial deployment potential.

1. Introduction

1.1. Background introduction

Printed Circuit Board (PCB) serve as the core components supporting and connecting electronic components, finding widespread application in consumer electronics, industrial control, aerospace, and other fields. As the performance and integration of electronic products continue to advance, PCB manufacturing processes grow increasingly complex. From substrate processing and component soldering to surface coating, defects can arise at every stage. During the post-printing soldering process, factors such as substrate material inconsistencies, solder joint contamination, and soldering iron temperature can cause five common soldering defects, as illustrated in Fig. 1: solder residue, dissymmetry, interconnect pad, aiguille, and holes. These defects not only impair circuit functionality but can also cause short circuits, overheating, and other issues, leading to significant economic losses (Zhong and Ma, 2022). Consequently, PCB surface defect detection has become a critical step in ensuring product quality and reliability.

1.2. Problem description

PCB surface defect detection utilizes human vision, laser scanning, industrial cameras, and other equipment to capture information about PCB surface traces or solder joints. It compares the characteristics of the circuit board under test against defect-free standards, identifying and classifying defects by recognizing discrepancies between the two. The probability of PCB surface defects occurring during soldering is influenced by multiple factors including material properties, process parameters, and molten solder behavior. Aiguille and interconnect pad defects are highly sensitive to overall production line process parameters; any parameter deviation can cause them to appear in large numbers across an entire batch. The occurrence probability of holes often fluctuates with material batches, while solder residue is more frequent on production lines with unstable cleaning processes. Dissymmetry is often linked to the design rationality of specific components, with relatively scarce sample data. Combined with the random nature

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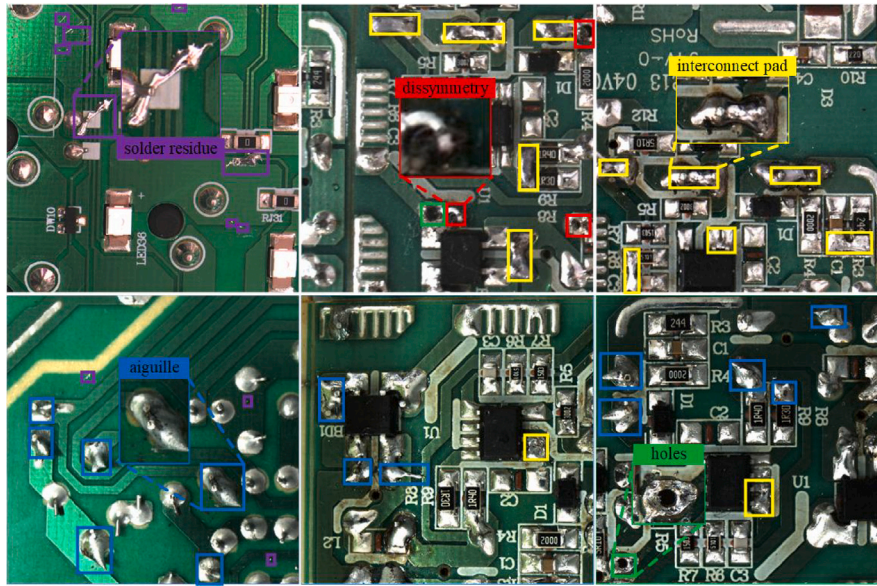


Fig. 1. Schematic diagram of PCB surface soldering defects.

of defect occurrence, this leads to a long-tail distribution of defect counts in actual production. Furthermore, the physical origins of each defect result in significant size variations. Aiguille represents minute threading formed during solder shrinkage, constituting a minute target. Interconnect pad defects occur at sizes at least multiples of the pad pitch, appearing in images as large connected abnormal structures. Solder residue exhibits a broad size range, making prediction challenging. These factors contribute to the long-tail distribution in defect quantity, dimensional diversity, and the microscopic nature of certain defects during PCB inspection. This complexity significantly increases the difficulty of accurately identifying and classifying surface defects on PCBs.

The PCB surface defect detection industry employs diverse inspection technologies, yet these face inherent limitations. Visual inspection by the human eye (Kim et al., 2021) heavily relies on the experience of operator and condition, resulting in low detection efficiency. Automatic Optical Inspection (AOI) (Deng et al., 2018) technology, which utilizes image analysis, has improved detection efficiency but suffers from poor robustness, making it difficult to adapt to complex and variable inspection scenarios. The advent of machine learning (Khan et al., 2025b,d) has boosted industrial inspection efficiency. Machine learning approaches utilizing feature engineering and classifiers (Hua et al., 2018) have been applied across various defect detection scenarios, but challenges remain, including strong sample dependency and limited adaptability to dynamic defects. While these traditional detection techniques possess certain defect detection capabilities, the accuracy in PCB inspection where foreground and background similarity is high can diminish due to operator fatigue or environmental influences. The development of deep learning has provided technical support for detection (Khan et al., 2025a,c). Image-based defect detection methods, which train networks using multi-threaded approaches (Khan et al., 2024) to autonomously learn defect features, then infer new images to output defect locations, types, and confidence levels, are widely adopted. Hybrid dual-backbone networks (Deng et al., 2024) enhance feature extraction capabilities by integrating auxiliary and true backbones in parallel; Dual convolutional kernels in YOLOv8n optimize backbone architecture to enhance defect feature representation and recognition capability (Luo et al., 2024). Methods like Mosaic data augmentation and adaptive anchor frame computation improve backbone structure (Zhang et al., 2024b), strengthening connections between shallow and deep layers to boost PCB defect detection performance.

Although backbone optimizations enhance defect recognition, low-frequency defects are often neglected, leading to insufficient information exchange between low-frequency features and multi-level features. FPN (Lin et al., 2017a) employs a multi-scale feature pyramid to superimpose shallow and deep features. TDD-Net (Ding et al., 2019) achieves multi-scale feature fusion through simple cross-layer feature concatenation. Improved Faster-RCNN (Hu and Wang, 2020) introduces top-down upsampling and lateral connection strategies to integrate multi-level features. LDD-Net (Zhang et al., 2024a) designed a Multi-Scale Aggregation Network (MAN) to promote information sharing between feature maps at different scales. Although diverse feature fusion methods have been employed to integrate multi-level features and enable interaction between shallow and deep features, most approaches rely on static weighting strategies. Such reliance hinders dynamic adjustment of fusion weights according to the actual scale of input samples, thereby limiting adaptability to multi-scale defects. Spatial deformable convolutions (SDC) (Huang et al., 2024) boost multiscale feature extraction for complex defects; while the method combining deformable convolutions with region of interest (ROI) alignment (Song et al., 2023) improves detection performance for large-scale, complex, and irregularly shaped defects. While employing convolutional kernels of varying sizes mitigates the impact of scale differences on detection accuracy, the continuous downsampling during network propagation leads to feature loss in small objects, thereby limiting the capability for detecting minute targets.

In summary, although existing models perform well in general scenarios, limitations remain in detecting PCB surface defects characterized by long-tail distributions, significant size variations, and minute targets. First, when defects exhibit a long-tail distribution, tail-end categories suffer from sparse samples and insufficient feature representation capabilities. During training, models tend to favor head categories with more defects, resulting in constrained overall detection accuracy. Although data augmentation can mitigate sample imbalance, such strategies often overlook the practical need for detecting rare defects in production, failing to fundamentally enhance model generalization. Furthermore, when defects vary significantly in size, multi-scale feature fusion methods using static weighting strategies struggle to dynamically adjust fusion weights based on the actual scale of input defects, limiting the model's ability to extract features from multi-scale defects. Simultaneously, constrained by the planar receptive field of conventional two-dimensional (2D) convolutional kernels, the model struggles to effectively perceive defect feature variations with

depth information, leading to frequent missed detections and false positives. Finally, the loss of detail in minute features of micro-PCB surface defects during propagation due to resolution degradation across layers combined with the limited capture capability of traditional inspection head structures for fine features, further constrains the performance of model in locating and classifying minute defects.

1.3. Research objectives

To achieve multi-scale defect detection on PCB surfaces in long-tail small-target scenarios, a Long-Tail Dynamic Multi-Scale Printed Circuit Board (LDM-PCB) is designed in this paper based on image processing. The Long-Tail Feature Extraction Network (LTFE-Net) is proposed, centered on the Adaptive Tail Attention (ATA) module. Through the utilization of the AttnStar C2f architecture, cross-level detail enhancement and feature reconstruction are achieved. When combined with the ADown module, precise localization of defect regions is enabled, enhancing the representation capability of tail defects. The Dynamic Multi-Scale Fusion (DMS-Fuse) architecture incorporates an adaptive weight allocation mechanism by which fusion weights are dynamically adjusted based on the large, medium, and small size characteristics of input defects. Both global and detailed information of defects across different scales are captured. Concurrently, depth information of defects is captured through 3D convolutional operations, effectively enhancing the model's perception of multi-scale defects. A Dynamic Prediction Layer (Dy-P2) is designed. By preserving high-resolution features, defect details are retained and deep semantic information is dynamically fused across layers. Through this layer, the rich fused semantic information is directly output to the detection head, improving the localization accuracy and feature retention capability for minute defects. Key contributions are summarized as follows:

(1) To address the long-tail distribution of defect data, the Long-Tail Feature Extraction Network (LTFE-Net) is proposed. Centered on the Adaptive Tail Attention (ATA) module, the AttnStar C2f architecture is employed for cross-level detail enhancement and feature reconstruction. When combined with the Adaptive Downsampling Convolution (ADown) module for precise defect localization, feature learning imbalance under long-tail data distributions is mitigated.

(2) The Dynamic Multi-Scale Fusion (DMS-Fuse) architecture is developed to address the limitation of fixed feature fusion weights in adapting to multi-scale defect representations. In this architecture, fusion weights are dynamically adjusted based on input defect characteristics, while 3D convolutional operations are incorporated to capture depth-information-rich defects, thereby enhancing fusion and perception capabilities for multi-scale defect features.

(3) To address feature vanishing of minute defects in deep networks, the designed Dynamic Prediction Layer (Dy-P2) leverages high-resolution retention and dynamic cross-layer semantic collaboration mechanisms. This optimizes multi-scale object detection robustness while enhancing fine-grained defect capture capabilities.

(4) The Long-Tailed PCB dataset is constructed, comprising 940 images across five defect categories. This dataset is characterized by an imbalanced defect distribution, simulating the random occurrence of defects in real detection scenarios. Quantitative and qualitative experiments are conducted using this dataset to evaluate the performance of the proposed method. Strong generalization capability is further validated on the PKU-Market-PCB augmented dataset (Ding et al., 2019; Feng and Cai, 2023).

The structure of this paper is organized as follows: Section 2 reviews advanced methods in the field of target defect detection; Section 3 describes the theoretical framework and algorithm implementation of the proposed novel detection method; Section 4 validates the effectiveness of the proposed method through ablation studies, multi-group comparative experiments, generalization experiments, and edge deployment experiments; Section 5 summarizes the contributions and outlines directions for future research. LDM-PCB Inspection Flowchart as shown in Fig. 2.

2. Related work

2.1. Object detection algorithms based on convolutional neural networks

Although existing methods have made progress in the areas of long-tail data optimization, multi-scale feature fusion, and tiny target detection in the field of defect detection, there are still limitations. In addressing the issue of class imbalance between positive and negative samples, Focal Loss (Lin et al., 2017b) can be employed to suppress the loss contribution from samples that are easily classified, thereby enhancing detection performance. Nevertheless, it remains challenging to effectively balance feature representation for tail categories within long-tail data distributions. OHEM (Shrivastava et al., 2016) is a technique employed in the context of online hard example mining to enhance the learning of tail defects. However, it has been demonstrated that excessive reliance on noisy samples has the potential to result in model overfitting. Methods that strengthen cross-channel feature extraction and enhance multi-scale feature fusion (Gao et al., 2025) improve detection accuracy for minute defects in photovoltaic cells, yet their approach neglects tail defects and fails to account for the impact of tail data on detection precision. Different detection models built on pre-trained VGG16 convolutional neural networks (Golding et al., 2022) addressed the low detection efficiency of traditional methods for concrete cracks. However, this study did not account for the actual long-tail distribution characteristics of cracks. UAV-YOLOv8 (Wang et al., 2023a) has been developed to design an efficient feature fusion module to enhance object detection performance. Nevertheless, it does not adopt a dynamic fusion strategy during feature integration. BL-YOLOv8 (Wang et al., 2023b) introduced the SimSPPF module to replace the traditional SPPF. Although it enhances the neck feature fusion efficiency and improves the detection accuracy, the design comes at the cost of losing multi-scale information, which leads to the loss of multi-scale features. Although improved deep learning methods for automatically detecting rail surface defects (Faghih-Roohi et al., 2016) demonstrate significantly higher detection accuracy than traditional models, these approaches typically employ sliding windows for defect localization. Determining optimal window sizes becomes challenging when input images vary in size and aspect ratio. By refining loss functions and integrating compression techniques (Liu et al., 2025), accuracy was enhanced while reducing parameters on long-tail datasets of turbine blade surface defects. However, room for improvement remains in detecting small objects. The improved NMS algorithm (Cao et al., 2019) prevents the loss of overlapping small targets but overlooks the difficulty in recognizing small targets in low-resolution images. DDNet (Yu and Wang, 2025) employs dense skip connections to enhance feature fusion for defects of various sizes on used book surfaces, improving model accuracy on self-built long-tail datasets. However, the model's overall detection accuracy remains low due to the impact of tail defects.

2.2. Object detection algorithms based on transformers

In addition to traditional convolutional neural network (CNN) architectures, Transformer-based frameworks have shown potential in industrial defect detection scenarios in recent years. Although Transformer-YOLO (Pan et al., 2025) utilizes the Swin Transformer architecture to improve defect detection accuracy, it fails to take into account the inter-class imbalance of defects. Although the ETDNe (Zhou et al., 2023) detection network improves the characterization of global defect features by introducing a Transformer, this strategy can lead to biased results when dealing with long-tailed training data. LSwin Transformer (Zhu et al., 2023) lacks in-scale feature interaction capability in defect detection and is unable to capture 3D information about defects. MSFT-YOLO (Guo et al., 2022) added a TRANS module based on Transformer design to the backbone network and detection header, which effectively enhanced the global information fusion capability,

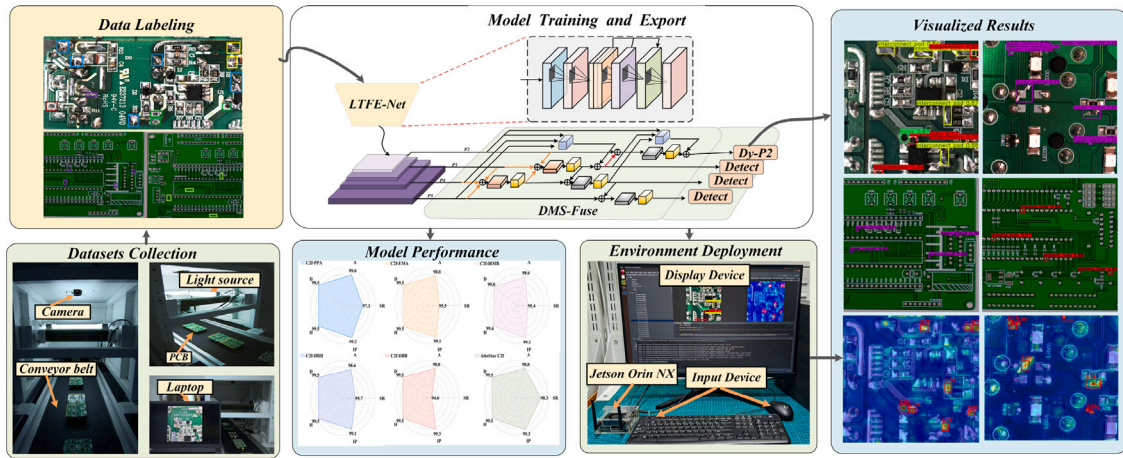


Fig. 2. LDM-PCB Inspection Flowchart.

meanwhile, the overall computational volume of the model increased, which also affected the real-time performance of detection. Cas-VSwIn Transformer (Gao et al., 2022) enhances defect detection, albeit at the cost of disregarding semantic information concerning diminutive targets. This enhancement is facilitated by an optimized window-shifting mechanism and a feature fusion strategy.

2.3. Object detection algorithm for PCB defect detection

As an important branch of industrial defect detection, PCB surface defect detection has led to the development of many methods with both exploratory value and practical significance. For example, the Faster R-CNN network (Hu and Wang, 2020), which applies ResNet50 as the backbone for feature extraction, improves the detection accuracy of small defects on PCBs. A new multi-scale feature pyramid network (Zhu et al., 2025) has been proposed to enhance the fusion of contextual PCB defect information. The improved YOLOv7 network (Yang and Kang, 2023) adds a convolutional layer to extract small PCB features. The Rev-RetinaNet network (Tang et al., 2023), based on RetinaNet and using ConvNext as the backbone, improves the detection accuracy of small PCB defects. The construction of LPViT, a small-size PCB inspection network based on a Transformer, improves the accuracy and robustness of the inspection (An and Zhang, 2022). Transformer-YOLO (Chen et al., 2022) leverages the Swin Transformer architecture to enhance defect attention capabilities, thereby improving PCB defect detection accuracy. As shown in Fig. 3, although current detection methods have improved defect recognition capabilities to some extent, the inherent severe imbalance problem in PCB surface defects remains inadequately addressed. Furthermore, critical challenges persist, including the multiscale variation of defect features and the loss of contextual information due to continuous downsampling operations. These issues are particularly pronounced in PCB surface defect detection under complex backgrounds.

3. Methods for circuit board defect detection

This paper employs YOLOv8 (Varghese and Sambath, 2024) as the baseline model and constructs the Long-Tail Dynamic Multi-Scale Printed Circuit Board (LDM-PCB) architecture for defect detection based on image processing, as illustrated in Fig. 4. The Long-Tail Feature Extraction Network (LTFE-Net) constructs high-dimensional feature maps to effectively enhance the detailed characteristics of low-frequency defects. This strengthens the model's ability to represent tail defects in long-tail distributions, enabling more precise capture of rare defect details. The Dynamic Multi-Scale Fusion (DMS-Fuse) architecture adaptively adjusts feature fusion strategies based on the scale characteristics of input defects. By utilizing 3D convolutions to extract defect

depth information, it enhances the model's perception and representation capabilities for multi-scale defects. The Dynamic Prediction Layer (Dy-P2) preserves high-resolution defect features while establishing an effective cross-level feature interaction mechanism. By directly outputting dynamically fused information, it mitigates detail degradation in deep networks and improves the localization accuracy of subtle defects. This method effectively addresses challenges such as long-tail distributions, cross-scale defect recognition, and the optimization of small feature collaborative capture in industrial scenarios.

3.1. The backbone feature extraction network LTFE-Net

In PCB surface defect detection tasks, defect categories often exhibit an abundance of high-frequency head defect samples and a scarcity of low-frequency tail defect samples. Although existing feature extraction methods possess a certain degree of cross-scenario generalization capability, when models encounter long-tail data distributions, their feature extraction processes tend to be dominated by head defect categories. This weakens the model's ability to recognize tail defects. To address the issue of insufficient learning of tail features for PCB defects, and inspired by the works in Ma et al. (2024), Wang et al. (2024b), and Cai et al. (2024), this paper designs the LTFE-Net to enhance the capability of a network to capture tail features of PCB defects. As shown in Fig. 5, LTFE-Net employs an Adaptive Tail Enhancement (ATA) module at its core, utilizing feature maps extracted via P1 cross-pixel convolution as input. By constructing a high-dimensional feature space, it enhances the representation capability of tail defects, making them easier to identify. Parallel pooling operations enable rapid localization, while an attention mechanism strengthens focus and responsiveness to tail defect features, thereby improving the detection of network performance for tail data.

3.1.1. ATA module

The proposed ATA module consists of a cascaded ADown and AttnStar C2f architecture. Adaptive Downsampling (ADown) selectively preserves critical tail defect information and detailed features by enhancing the downsampling process, mitigating feature loss during downsampling. The AttnStar C2f module effectively integrates shallow texture details with deep semantic information within the backbone network through element-wise multiplication of cross-layer PCB feature maps. This constructs an information-rich composite feature space, amplifying the feature response of tail defects to enhance their detectability and localization in complex backgrounds. Simultaneously, the embedded Contextual Anchor Attention (CAA) mechanism enhances the network's global focus on tail defect features. AttnStar C2f collaborates with the ADown module to achieve complementary optimization. This approach maximizes the retention of critical information for tail PCB defects while effectively reducing model parameters and improving detection speed.

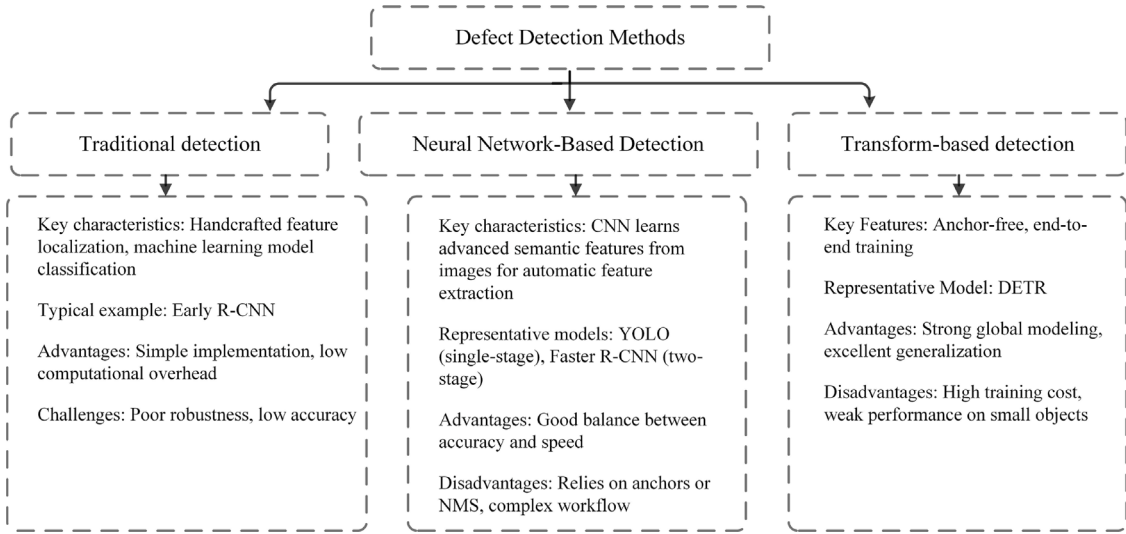


Fig. 3. Classification of defect detection methods.

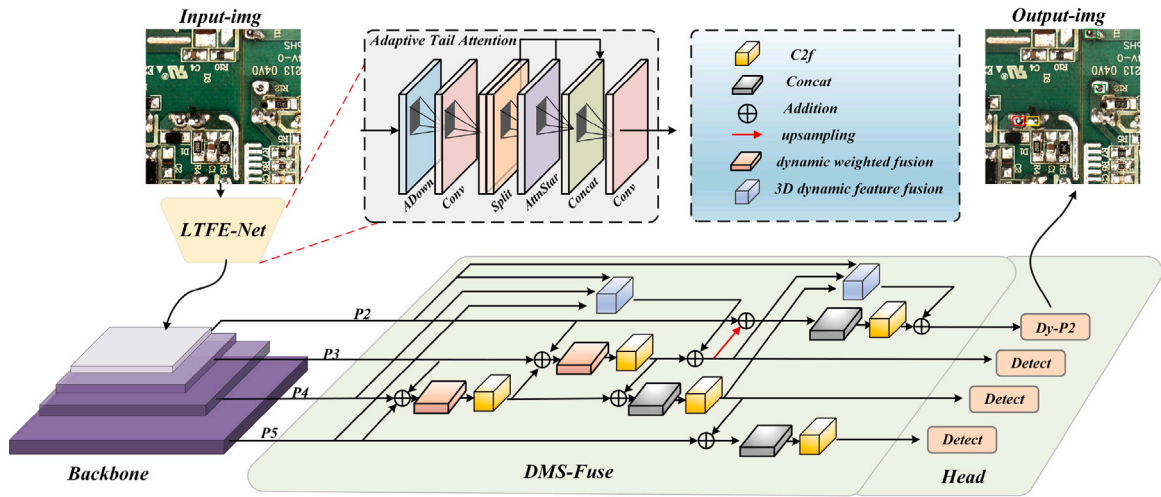


Fig. 4. The architecture of LDM-PCB.

3.1.2. ADown module

In PCB defect detection tasks based on convolutional neural networks, convolutional operations are typically applied to input images to extract defect-related local features while reducing computational costs. However, traditional standard convolutions are constrained by their local receptive fields, and the convolutional process often leads to gradual attenuation of feature information. The backbone network of baseline models performs successive downsampling on input images, causing PCB defect feature information to degrade layer by layer. This is particularly pronounced when defects exhibit long-tail distributions, where feature responses from tail defects become progressively diluted through multiple downsampling iterations, weakening the model's ability to identify low-frequency defects. ADown employs a bidirectional parallel architecture to simultaneously execute average pooling and max pooling operations. Average pooling rapidly filters out irrelevant background regions, while max pooling enhances activation and localization capabilities for local defect features. This hybrid pooling strategy reduces redundant computations in defect-free areas while selectively preserving critical semantic information and detailed features, mitigating the loss of tail-end feature information during the downsampling process.

3.1.3. AttnStar C2f module

To address the limitations of traditional C2f methods—which rely on cross-pixel convolutions and feature stacking, resulting in insufficient response to tail defects and loss of feature information—the AttnStar C2f module constructs a high-dimensional implicit feature space through element-wise multiplication and attention-cooperative mechanisms. This amplifies defect features and enhances the expression capability of PCB defect characteristics. CAA enhances the network's global capability to capture long-tail defect details and its feature-focusing ability by learning attention weights across channel dimensions, enabling the network to concentrate on channels containing tail defect features. Centered on the AttnStar mechanism shown in Fig. 6, AttnStar C2f integrates star operations (Ma et al., 2024) with CAA to form a progressive optimization chain for global feature enhancement and tail-end feature focusing. Its characteristics are described by Eq. (1):

$$F_{\text{fuse}} = \phi(F_1 * F_{\text{DW}}) \odot (F_2 * F_{\text{DW}}) \quad (1)$$

where ϕ represents the ReLU activation function, F_1 and F_2 are parallel branches, $\odot$ denotes element-wise multiplication, and the 1×1 convolution adjusts the dimensions and channels of the image, and

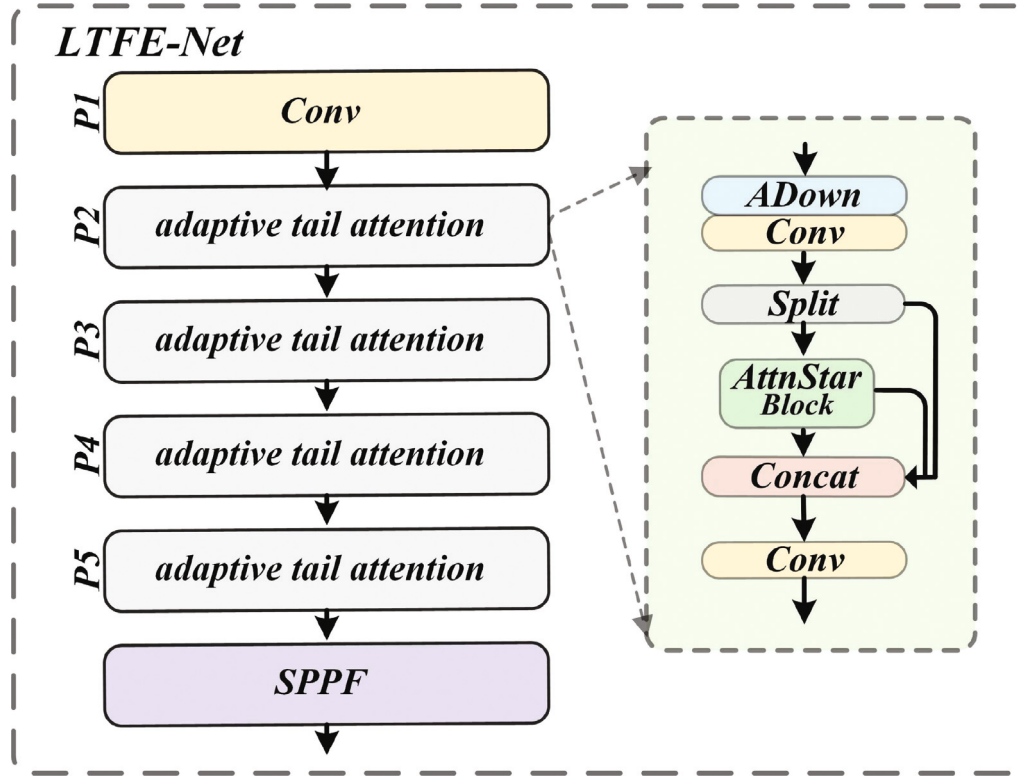


Fig. 5. Schematic diagram of LTFE-Net structure.

introduces nonlinear interactions in the feature space. The introduction of element-wise multiplication operations in inter-layer feature maps effectively promotes interaction and fusion among multi-level features, thereby more comprehensively preserving critical defect characteristics. Furthermore, by implicitly modeling high-dimensional defect regions, it enhances the local feature representation of tail defects, effectively mitigating the vanishing of fine details in deep networks. Meanwhile, CAA (Cai et al., 2024) adopts a channel-optimized-space-adjusted single-pass design, the principle of which can be characterized by Eq. (2):

$$F_{\text{attn}} = A_C(F_{\text{fuse}}) \otimes A_{\text{DW}}(F_{\text{fuse}}) \quad (2)$$

where, A_C generates the channel-weight vector through global average pooling and a fully-connected layer, helping to suppress noise interference from redundant channels. Each convolutional kernel in A_{DW} performs convolutional operations with only one of the corresponding channels, which enables the tail features to be adjusted in spatial dimensions, so that the network can focus on the tail defect region more accurately, and assists ADown to achieve adaptive capture of tail defects. The AttnStar C2f module enhances the ability to characterize and localize long-tail defects in networks without significantly increasing computational complexity, demonstrating superior adaptability particularly when detecting subtle defects.

3.2. Dynamic multi-scale feature fusion framework

In object detection networks, feature interaction across scales is crucial, especially in the field of industrial PCB inspection. The large variation in defect sizes, uneven distribution of defect types, and complex background interference make existing detection networks limited in handling multi-scale feature fusion and three-dimensional depth perception. The commonly used feature fusion network adopts a fixed-layer sampling strategy, which is prone to lose small defect details during cross-scale feature alignment, and the static fusion mechanism is unable to fully capture effective defect features. Additionally, the

2D convolution in feature fusion networks struggles to understand the depth gradient features of different defect deformations on the PCB surface, which leads to the leakage of defects relying on depth information including interconnect pad and solder residue. To better achieve the fusion and perception of key PCB features, the DMS-Fuse framework proposed in this study addresses the issues of information loss and redundancy in multi-scale feature fusion through adaptive weighted fusion and dynamic scale alignment strategies. By replacing 2D convolutions with 3D convolutions, it enhances the 3D perception ability of PCB defects. As shown in Fig. 7, a dynamic weighted fusion strategy is designed for features at different scales. For large-scale features, weighted adaptive pooling is applied to preserve both global information and local details, while for small-scale features, bilinear interpolation is employed for upsampling to ensure spatial consistency across different scales. After alignment, all features are standardized to eliminate statistical bias, and the weighted fusion (Lin et al., 2017b) is represented by Eq. (3):

$$F_{\text{out}} = w_l \cdot l' + w_m \cdot m + w_s \cdot s' \quad (3)$$

The equation, l' represents the output of the large-scale features after weighted pooling, m is the medium-scale feature, and s' is the output of the small-scale features after upsampling. w_l , w_m , and w_s are the dynamic weight coefficients for the large-scale, medium-scale, and small-scale features, respectively, and satisfy the condition $w_l + w_m + w_s = 1$.

Unlike the fixed multi-scale fusion approach employed in traditional YOLO variants, the dynamic weighting fusion strategy can flexibly adjust the weights of features at different scales based on the specific variations in the input features. When detecting small defects, the network enhances the weight coefficients of shallow feature maps to highlight the detailed information in the defects. When detecting larger-size defects, boosting the fusion weights of the deeper semantic features is more effective at capturing the global information. This flexible feature fusion effectively improves the perception accuracy of the model and robustness to multi-scale defect features, which in turn

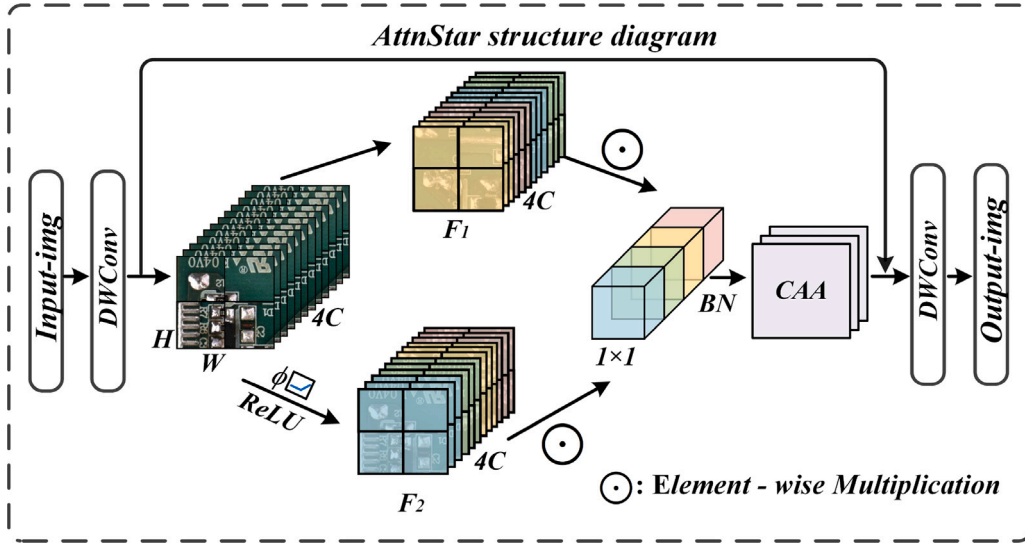


Fig. 6. The AttnStar structure schematic diagram.

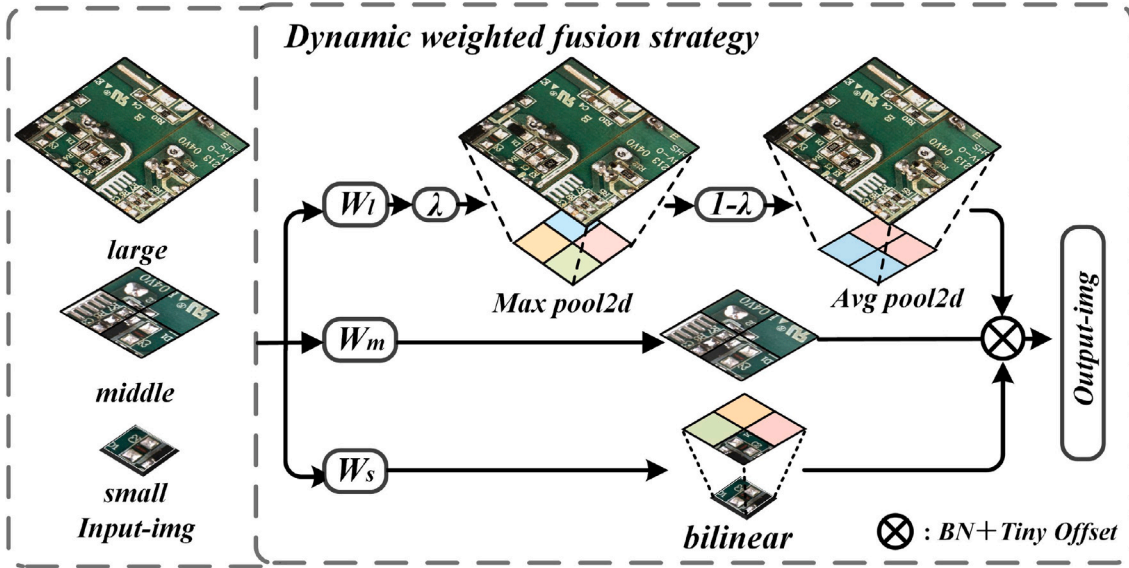


Fig. 7. Dynamic weighted fusion strategy.

helps the network accurately capture the different characteristics of tiny and large-size defects on the PCB surface.

As shown in Fig. 8, to address the 3D spatial features of PCB defects, including surface deformation and texture anomalies, a 3D convolution kernel is employed to construct the depth perception network architecture, extending the traditional 2D convolution. In Eq. (4), high-resolution shallow features F_s , mid-level features F_m , and large-scale features F_l are spatially aligned, through dynamic sampling (Liu et al., 2023), retaining the planar dimension information. After aligning multi-scale features, 3D convolution is applied through sliding convolution operations along the depth dimension to effectively extract the 3D deformation features of defect regions. The 3D feature interaction mechanism can effectively model the 3D geometric distortion information of defects such as pad adhesion and solder residue, which improves the defect characterization capability with complex spatial structure compared to the traditional 2D convolution and improves the ability of the network to sense the depth of defects.

$$F_{\text{fused}} = C_{3D} \otimes \text{Concat}(F_l, \text{DySample}(F_s), \text{DySample}(F_m)) \quad (4)$$

3.3. The head section designs the Dy-P2 structure

To address the problem of resolution limitations in PCB feature layers leading to the loss of micron-level defect details, and the inability of the static feature fusion mechanism to dynamically adapt to multi-scale defect features, this study proposes the structure of the Dy-P2 layer based on a multi-level dynamic synergy mechanism, as shown in Fig. 9. Through a synergistic approach of resolution preservation and semantic interaction, this architecture is designed to enhance the detection of minute, millimeter-scale targets, specifically such as aiguilles and small holes. The Dy-P2 layer adopts a progressive downsampling strategy: it performs only two downsampling operations on the backbone network, allowing the shallow feature maps to retain high resolution at 1/4 of the original input size, thus improving spatial detail retention by a factor of four compared to typical detection networks. Additionally, the DMS-Fuse architecture enables dynamic semantic propagation across layers, facilitating cross-scale interaction between deep network-extracted large receptive field features and shallow high-resolution

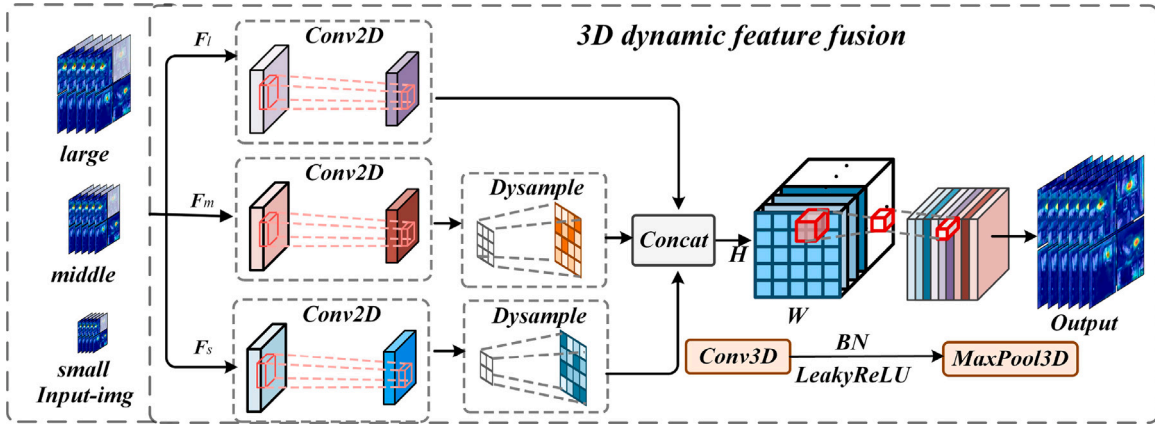


Fig. 8. The 3D dynamic feature fusion.

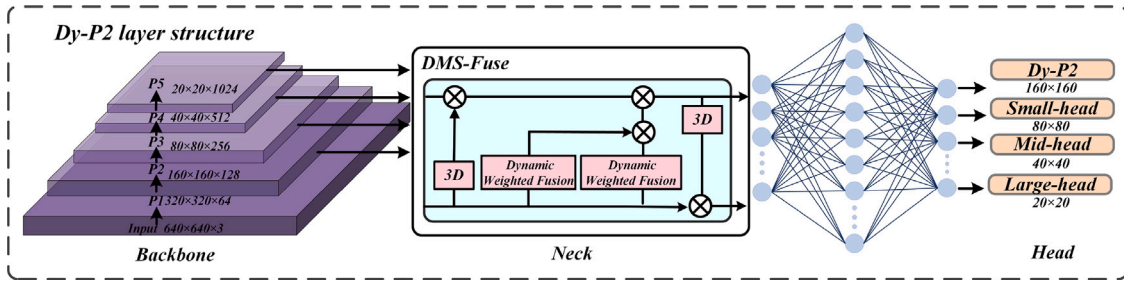


Fig. 9. The Dy-P2 layer structure.

features. The interactive information is directly output to the detection layer, effectively resolving the spatial-semantic conflict of multi-scale features. This multi-scale dynamic collaboration design allows the Dy-P2 layer to handle small targets while enhancing the perception of larger targets, thereby improving the performance of the model in complex environments. Compared to the traditional P2 layer, the Dy-P2 layer enables the network to better capture the detailed features of small targets, improving prediction performance for small defect targets.

4. Experimental validation

This chapter systematically validates the practicality and generalization capability of the methods proposed in Chapter 3 through experiments. The specific structure is as follows: Section 4.1 details the acquisition and construction process of the self-built Long-Tailed PCB dataset; Section 4.2 explicitly lists the hardware configuration, software environment, and parameter initialization settings required for the experiments; Section 4.3 introduces the model performance evaluation metrics adopted in this study; Section 4.4 conducts ablation experiments on the proposed innovative modules. By combining quantitative results with visual analysis, it validates the effectiveness of each module and further optimizes model architecture selection through comparative improvements of different C2f modules. Section 4.5 compares the proposed algorithm with current mainstream image-processing-based defect detection methods, demonstrating the effectiveness of our approach. Section 4.6 generates and compares prediction heatmaps of different models through visualization methods, intuitively demonstrating the proposed method's localization capability for defect regions. Section 4.7 conducts generalization experiments on public datasets to further validate the algorithm's broad applicability. Section 4.8 evaluates the algorithm's deployment performance in edge computing environments based on the NVIDIA Jetson Orin NX embedded platform.

Table 1

The camera parameters during data collection.

Project	Parameter
Camera model	MV-CS060-10UC-PRO
Aperture	F4
Focal length	8 mm
Pixel size	2.4 μm $\times$ 2.4 μm
Exposure time	50000 μs
Camera resolution	3072 $\times$ 2048
Maximum frame rate	59.6 FPS
Signal-to-noise ratio	41.3 dB

4.1. Construction of the Long Tailed PCB dataset

Fig. 10 displays the field equipment for PCB data acquisition. Experimental data in this study was captured using Hikvision industrial cameras, targeting surface soldering defects on Printed Circuit Boards measuring 58 mm $\times$ 31 mm. The test samples featured over 30 electronic components and more than 70 solder joints.

The image dataset comprehensively covers five typical types of soldering defects: aiguille, interconnect pad, dissymmetry, holes, and solder residue. The relevant camera parameters for image collection are detailed in Table 1.

As shown in Fig. 11, various defects can occur during PCB manufacturing. Although manifesting differently, all these defects impact product quality. Aiguille are structurally fragile, elongated metallic protrusions that form on the surface or edge of a solder joint, which can lead to fracture failure and a high risk of electrical short circuits. The interconnect pad presents as an unintended conductive bridge formed between adjacent pads, which could lead to impedance anomalies and current leakage. Dissymmetry refers to the asymmetric distribution or positional shift of solder on the joint interface, directly affecting the mechanical strength and structural reliability of the solder joint. Holes refer to cavity-like structures formed at the solder interface or within

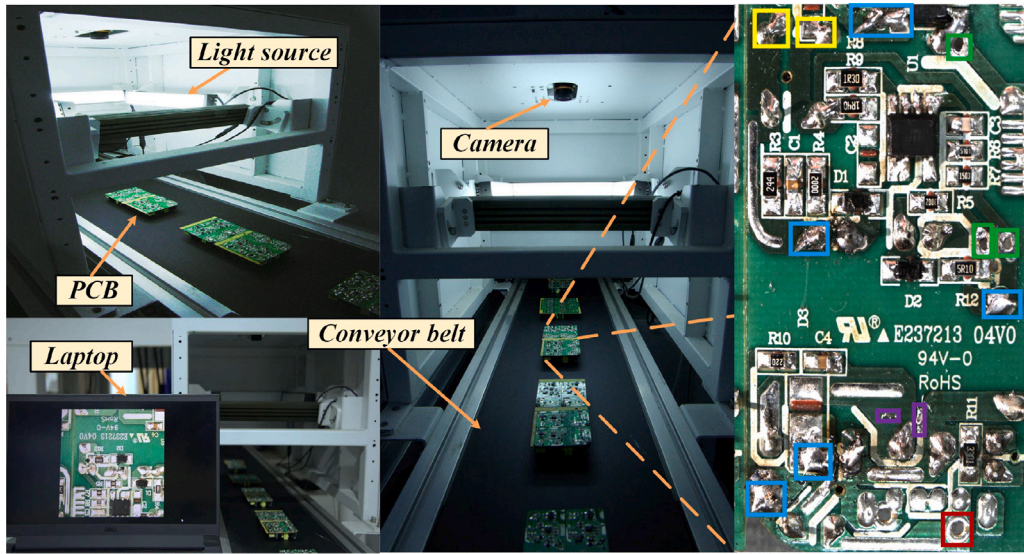


Fig. 10. Image acquisition equipment.

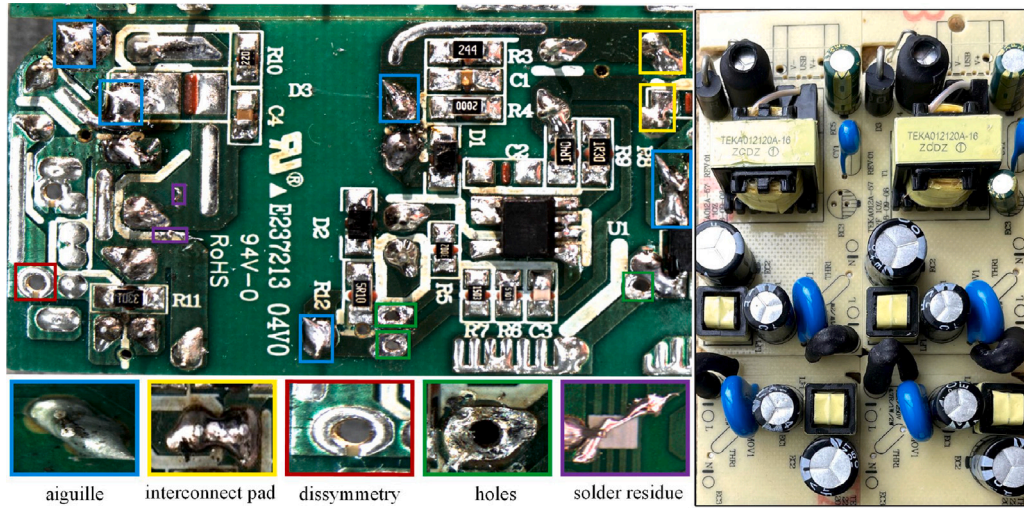


Fig. 11. PCB surface defect images.

the joint, which weaken electrical conductivity and accelerate thermo-mechanical fatigue. Solder residue refers to metallic micro-particles or solder paste deposits remaining in non-target areas, which are able to degrade the insulating properties of the dielectric layer.

Due to the limited number of original defect images collected, it is challenging to train a reliable and stable model. This section first employs a sliding crop method to augment the data from 250 original defect images. With a sliding step size of 320 pixels, a total of 940 defect images measuring 640×640 pixels were ultimately obtained. Subsequently, three annotators trained under a unified annotation protocol manually labeled five defect categories within the images. The specific category and quantity distributions were statistically analyzed, and the constructed dataset was named “Long Tailed PCB”. It should be noted that to maintain consistency with the original acquisition scenes as much as possible, no additional augmentation strategies were introduced beyond sliding window cropping for data expansion. A detailed comparison of the data before and after augmentation is shown in Table 2.

As observed, the sample proportion and quantity of the dissymmetry and solder residue defect categories are significantly lower than those of other categories. This differs from the balanced defect type

Table 2

Comparison table of defect quantities before and after data augmentation.

Defect category	Original defect quantity	Defect quantity after augmentation
aiguille	833	1719
dissymmetry	464	492
holes	701	1198
interconnect pad	563	1247
solder residue	796	720

distribution in traditional datasets (Ding et al., 2019; Huang and Wei, 2019) and better reflects the randomness of defect types in industrial inspections, so as to improve the generalization ability of the model. The dataset was then randomly divided into training, validation, and test sets, containing 830, 110, and 110 images.

4.2. Experimental setup and parameter configuration

The models in this section's experiments were run on an Ubuntu 20.04.6 machine configured with torch2.0.0 and torchvision0.15.1. The hardware includes a 12th Gen Intel® Core™ i7-12700F 2.10 GHz

Table 3
Initial parameters of the training model.

Parameter	Value
Optimizer	SGD
Momentum	0.937
Initial learning rate	0.01
Learning decay rate	0.005
Number of training iterations	300
Input image size	640 × 640
Batch size	4
Max number of computer threads	4

CPU, 32 GB RAM, and an NVIDIA GeForce RTX 3090 Ti 24 GB GPU. To ensure fairness and reproducibility, all models utilized identical training, validation, and test datasets. No pre-trained weights were employed, and training was conducted with equally optimized hyperparameters. Input images were uniformly resized to 640 × 640 pixels, and Mosaic data augmentation was introduced to enhance generalization capabilities. Detailed parameter configurations are presented in Table 3.

4.3. Evaluation indicators

The PCB surface defect detector trades off between accuracy, parameter quantity, and detection speed. High-precision models usually have high computational complexity and are difficult to meet the requirements of detection frame rate; while lightweight and simple detection models pursuing high FPS may lose some accuracy and affect the accuracy of defect recognition. To comprehensively evaluate the model, this study constructs an evaluation system from three aspects: detection accuracy, model efficiency, and result stability.

4.3.1. Detection accuracy metrics

Detection accuracy is a core indicator for measuring the recognition ability of the model. In this study, the mean Average Precision (mAP) is used as the core evaluation criterion, and its calculation is built layer by layer based on Precision, Recall, and Average Precision (AP).

Precision: Measures the proportion of true positives among the positive samples predicted by the model. The calculation formula is:

$$\text{Precision} = \frac{TP}{(TP + FP)} \quad (5)$$

Among them, TP (True Positive) represents the number of correctly predicted positive samples, and FP (False Positive) represents the number of negative samples incorrectly predicted as positive samples. A high precision means that the model has fewer false positives.

Recall: Measures the proportion of all true positive samples that are correctly predicted by the model. The calculation formula is as follows:

$$\text{Recall} = \frac{TP}{(TP + FN)} \quad (6)$$

Among them, FN (False Negative) represents the number of positive samples incorrectly predicted as negative samples. A high recall rate means that the model has fewer missed detections.

Average Precision (AP): To comprehensively measure the performance of the model at different recall levels, the area under the Precision–Recall curve (PR curve) is used to define AP. The calculation formula is:

$$AP = \int_0^1 P(R) dR \quad (7)$$

The integral value is approximated by interpolating or summing the PR curve. AP comprehensively reflects the model's ability in both high recall and high precision.

Mean Average Precision (mAP): For multi-category defect detection tasks, mAP is an important criterion for measuring the overall detection

accuracy of the model. Its calculation method is to take the arithmetic mean of the AP values of all defect categories:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (8)$$

Among them, N is the total number of defect categories, and AP_i is the average precision of the i th category. For further detailed evaluation, this paper evaluates mAP from two IoU threshold scales.

4.3.2. Model efficiency metrics

Parameter Quantity: Measures the complexity of the model and the ability to occupy computing resources. A model with fewer parameters has a more obvious advantage in deployment on memory-constrained embedded devices.

Frames Per Second (FPS): The number of images that can be processed and inferred per second on specific hardware. It is a decisive indicator of whether the model can be applied to real-time online detection systems.

4.3.3. Result stability metrics

To meet the requirements of industrial applications, the algorithm must be stable and reliable, and be able to resist the interference of random factors such as environmental noise and data fluctuations. **Mean Absolute Deviation (MAD):** MAD is used to measure the stability of the model in multiple repeated experiments. The calculation formula is:

$$MAD = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}| \quad (9)$$

Among them, x_i is the mAP result obtained from a single experiment, $\bar{x}$ is the average mAP of n experiments, and n is the number of repeated experiments. A smaller MAD value indicates that the results of multiple experiments have smaller fluctuations, higher concentration, and the model has stronger robustness and reliability.

4.4. Ablation experiment

4.4.1. Comparison of the accuracy performance of the baseline model after processing through each module

To validate the effectiveness of the proposed method, this section compares its performance with baseline models through ablation experiments. All experiments were conducted on identical training, validation, and test datasets. Models were retrained using the parameters specified in Table 3, with evaluation metrics defined by Eqs. (5) to (8). The experimental results are shown in Table 4, sequentially displaying the performance changes after progressively introducing the AttnStar C2f structure, the ADown module, and the Dy-P2 layer, as well as the overall performance of the complete module combination.

Table 4 shows that the baseline model YOLOv8-s achieves $mAP_{0.5}$ of 97.3% and $mAP_{0.5:0.95}$ of 76.2% at 640 × 640 resolution. After introducing the DMS-Fuse structure at the network bottleneck, the model gains stronger adaptive dynamic feature fusion capabilities. It effectively captures defect features with depth information, enhancing defect detection performance and boosting $mAP_{0.5}$ to 97.8% (+0.5%) with minimal increase in parameters. The Dy-P2 module facilitates interaction between deep features with large receptive fields and shallow high-resolution features, further improving small object detection performance with $mAP_{0.5}$ reaching 98.7% (+1.4%). Simultaneously, it directly outputs fused information to the detection layer, reducing parameters while optimizing redundant structures. The introduced AttnStar C2f module performs element-wise multiplication operations on cross-layer feature maps, replacing traditional cross-pixel convolutions. This constructs higher-dimensional defect features, enhancing defect representation, achieving $mAP_{0.5}$ of 97.8% (+0.5%), though parameter count increases to 11.82 M, raising model complexity. When cascading AttnStar C2f with ADown, pooling operations enable rapid filtering of

Table 4
Results of the ablation experiment.

Baseline	Schematic design				Image-size	$mAP_{0.5}$ (%)	$mAP_{0.5:0.95}$ (%)	Params (M)	FPS
	DMS-Fuse	Dy-P2	ADown	AttnStar C2f					
YOLOv8-s					640	97.3	76.2	10.61	172
	✓				640	97.8	77.4	10.78	152
	✓	✓			640	98.7	79.5	8.66	126
	✓		✓		640	98.0	77.8	9.72	156
	✓		✓	✓	640	98.0	77.4	10.74	91
	✓			✓	640	97.8	78.5	11.82	119
	✓	✓		✓	640	98.8	79.2	9.70	85
	✓	✓	✓	✓	640	99.1	81.6	8.61	100
					640				

invalid background regions. This enhances activation and localization of local anomalous features while reducing redundant computations in defect-free areas. Parameter count decreases to 10.74 M, with $mAP_{0.5}$ improving to 98.0%. When all modules operate synergistically, the model achieves optimal performance: $mAP_{0.5}$ reaches 99.1% (+1.8%), $mAP_{0.5:0.95}$ attains 81.6% (+5.4%), with parameter size compressed to 8.61 M—an 18.6% reduction from the baseline—while demonstrating robust resistance to background interference and stable defect localization performance. Although inference speed drops to 100 frames per second for the full module combination, detection accuracy takes precedence over real-time capability in high-precision scenarios like industrial PCB defect detection. Therefore, the complete module combination from the last row of Table 4 is selected as the final model, named the Long-Tail Dynamic Multi-Scale Printed Circuit Board Detection Model (LDM-PCB). Subsequent sections will compare and validate it against other advanced detection algorithms.

To enhance the persuasiveness of the ablation experiments, more intuitive and reliable evidence is provided, with its visualization results shown in Fig. 12. After introducing the DMS-Fuse structure into the baseline neck of networks region, this module dynamically adjusts multi-scale feature weights and replaces traditional 2D convolutions with 3D convolutions to model defect depth information. This enhances the model's ability to capture defects at different scales, though false negatives still occur for some small-target defects. The introduction of the AttnStar C2f module constructs high-dimensional composite defect features through per-pixel operations, amplifying defect region characteristics to facilitate easier defect localization. Its attention mechanism also enhances global information perception, reducing the likelihood of tail-end category defects being overlooked. However, this simultaneously amplifies category defect noise, increasing misclassification risks. The ADown module achieves critical localization of defect regions through bidirectional parallel pooling operations, improving detection speed. However, its limited receptive field may cause local false negatives. When cascaded, AttnStar C2f and ADown function as effective complements: the former amplifies defect features, while the latter enables precise localization, thereby synergistically enhancing defect identification efficiency. Dy-P2 enhances detection of minute targets by directly conveying shallow detail features and deep semantic information to the detection layer. However, this also renders the model overly sensitive to defects, leading to false positives. Cascading AttnStar C2f with Dy-P2 further improves detection performance, yet remains susceptible to background interference with inconsistent detection across local regions. Ultimately, the LDM-PCB model achieves an optimal balance between detection localization accuracy and interference resistance, delivering the best overall detection performance and visualization results.

4.4.2. Optimization of the LDM-PCB model

To efficiently analyze the impact of multiple design parameters on structural performance, the optimal design structure was selected (Belhocine et al., 2021). This experiment used YOLOv8-s-DMS-Fuse-Dy-P2-ADown as the baseline model to evaluate the effect of the non-C2f combination module on model accuracy. The experimental results are shown in Table 5.

Comparison experiments with the baseline model YOLOv8-s-DMS-Fuse-Dy-P2-ADown show that the performance of different C2f modules in the PCB defect detection task is closely related to their design goals and task suitability. As shown in the experimental data in Table 5, the C2f-PPA module improves the overall $mAP_{0.5}$ of the model to 98.9% through the multi-branch feature extraction and attention fusion strategy. However, the cross-domain fitness feature extraction process results in a limited level of detail localization accuracy; the C2f-EMA module, due to the low-rank reconstruction property of the expectation maximization attention mechanism, reduces the attention to unnecessary information, but the accuracy of tiny defect localization is insufficiently leads to only 62.5% $mAP_{0.5:0.95}$ in SR class; C2f-iRMB and C2f-DRB modules are limited by the lightweight design and art style migration characteristics, respectively, and the overall $mAP_{0.5:0.95}$ is 77.3% and 78.0% in complex defect modeling; Although the C2f-DBB module achieves an advantage in IP class defect detection through its multi-branch structure, its redundant branches are not sensitive enough to the detection of long-tailed features resulting in a significant decrease in the accuracy of SR class detection. In contrast, the AttnStar C2f module proposed in this paper achieves optimal performance in both overall detection accuracy and adaptability to multiple defect types through the synergistic design of a star topology and a channel-spatial attention mechanism. As visually demonstrated by the defect detection accuracy comparison across different C2f models in Fig. 13, the AttnStar C2f module effectively mitigates the low detection accuracy of tail categories caused by imbalanced defect counts. Its superior performance further validates the structure's excellent adaptability to complex defect detection tasks.

4.5. Comparison with other methods

4.5.1. Quantitative comparison with other methods

PCB surface defect detection requires balancing accuracy, parameter count, and inspection speed: High-accuracy models involve complex computations, making it difficult to meet frame rate requirements; lightweight models prioritize high FPS but often sacrifice accuracy. In practical industrial scenarios, it is essential to ensure high accuracy while also considering inference speed and model lightweighting. To comprehensively evaluate LDM-PCB's multi-scale feature fusion and small object detection performance in long-tail distribution scenarios, this study selected advanced detection models from representative CNN and Transformer architectures as comparison baselines. The YOLO series, renowned for its excellent balance between accuracy, speed, and model complexity, has become a mainstream architecture in industrial inspection. Consequently, the latest variants from YOLOv9 to YOLOv12 were included. YOLOv9 incorporates the GELAN structure to enhance feature representation capabilities; YOLOv10 employs a decoupled head without post-processing to improve small object detection efficiency; YOLOv11 strengthens cross-layer feature fusion to optimize multi-scale perception; YOLOv12 focuses on lightweight edge-side design, offering greater practical deployment reference value. Concurrently, Transformer-based models (e.g., Swin Transformer, RT-DETR) leverage their global context modeling capabilities to offer a complementary perspective distinct from CNNs for identifying rare classes in

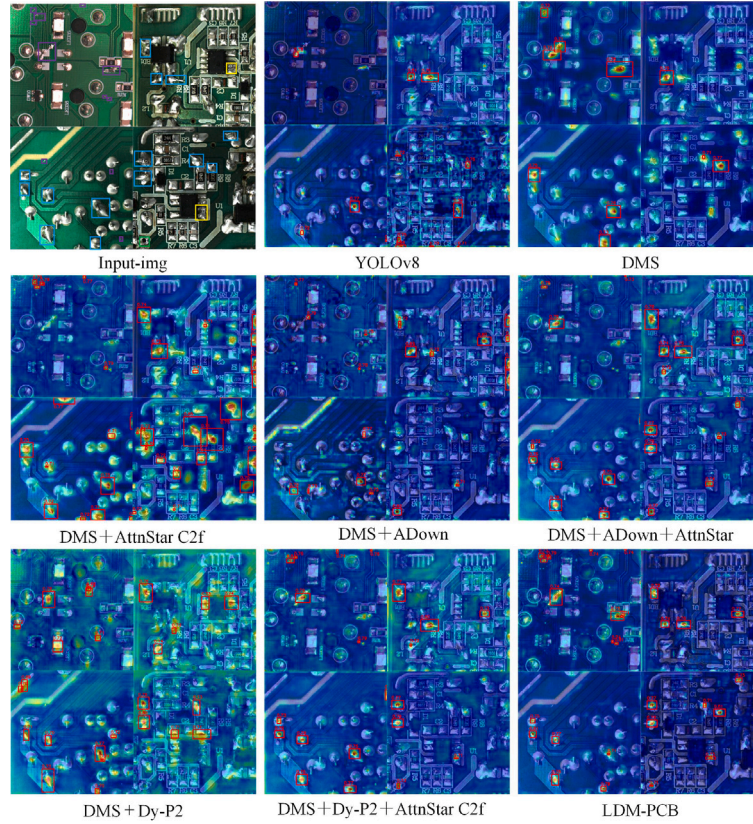


Fig. 12. Visualization of melting experiment results.

Table 5

Results compared to other methods. (In the experiment, solder residue defects are referred to as SR, aiguille defects as A, dissymmetry defects as D, holes defects as H, and interconnect pad defects as IP.)

Baseline	Module	$mAP_{0.5}$ (%)						$mAP_{0.5:0.95}$ (%)					
		SR	A	D	H	IP	ALL	SR	A	D	H	IP	ALL
YOLOv8-s-DMS- Fuse-Dy-P2- ADown	C2f-PPA (Xu et al., 2024)	97.3	99.0	99.5	99.5	99.2	98.9	65.6	87.8	84.9	72.8	83.9	79.0
	C2f-EMA (Li et al., 2019)	95.5	98.8	99.5	99.5	99.1	98.5	62.5	86.7	84.7	71.6	82.9	77.7
	C2f-IRMB (Zhang et al., 2023)	95.4	98.6	99.0	99.4	99.3	98.4	63.8	86.4	82.2	71.4	82.8	77.3
	C2f-DRB (Xu et al., 2021)	95.7	98.6	99.5	99.5	99.1	98.5	66.8	85.4	83.6	72.1	81.9	78.0
	C2f-DBB (Ding et al., 2021)	94.0	98.8	99.5	99.5	99.3	98.2	66.1	88.0	84.4	73.0	84.6	79.2
	AttnStar C2f	98.3	98.8	99.5	99.5	99.3	99.1	69.7	88.6	87.3	75.2	87.0	81.6

long-tail scenarios. All compared models were retrained on identical datasets using the parameters listed in Table 3 to ensure result comparability. The experiments encompassed 19 algorithms: Faster R-CNN, CenterNet, RetinaNet, EfficientDet-v1, YOLOX-s, YOLOv5, YOLOv8, the YOLOv9 to YOLOv12 series, and the proposed LDM-PCB model. Detailed results are presented in Table 6.

As shown in Table 6, LDM-PCB outperforms other methods on key accuracy metrics: The model achieves 99.1% $mAP_{0.5}$ and 81.6% $mAP_{0.5:0.95}$, with a MAD value of 0.08 in Eq. (9). It demonstrates stable and outstanding performance in locating long-tail complex defects and recognizing multi-scale minute semantic features. Although its parameter count slightly exceeds some lightweight models, its inference speed still meets industrial real-time demands, and the accuracy improvement holds practical value. Compared to LDM-PCB, the traditional two-stage detector Faster R-CNN exhibits insufficient attention to tail defects when processing long-tail distribution data, achieving only 49.9% $mAP_{0.5}$. Its region proposal mechanism also results in a high parameter count of 136.77 M. RetinaNet employs a fixed-scale feature fusion strategy, struggling to adapt to the diverse scales of PCB defects, resulting in a reduced $mAP_{0.5}$ of 76.0%. The heatmap-based CenterNet achieves fast inference speeds in keypoint detection but exhibits low sensitivity to minute defects and insufficient detail

capture, yielding a $mAP_{0.5:0.95}$ of only 46.9%. EfficientDet-v1 achieves model lightweighting through channel pruning and resolution scaling, but its detection accuracy is low, with a $mAP_{0.5}$ of only 61.3%, failing to meet high-precision application requirements. Swin Transformer achieves global context modeling via window attention mechanisms, attaining a $mAP_{0.5}$ of 96.8%, but its defect localization capability still needs improvement. RT-DETR employs a hybrid decoding mechanism to enhance detection accuracy for head categories, but its large model parameters result in high deployment costs. YOLOX-s strikes a good balance between accuracy and speed, achieving $mAP_{0.5}$ of 92.4% and FPS of 106, though its standard PANet architecture faces optimization bottlenecks in feature fusion. YOLOv5s achieves 217 FPS but requires improvement in identifying tail defects. YOLOv8s relies on anchor boxes, showing poor adaptability to irregular defects; its consecutive downsampling operations cause detail loss, resulting in a $mAP_{0.5:0.95}$ of 76.2%. YOLOv9s simplifies feature extraction, achieving $mAP_{0.5}$ of 97.2% while reducing model parameters by 1.8 M compared to LDM-PCB. However, this comes with minor loss of defect feature information. YOLOv10s employs a decoupled detection head and dynamic label allocation strategy, boosting FPS to 222. However, its detection capability for small-object defects weakens, with $mAP_{0.5}$ dropping to 96.4%. YOLOv11s enhances the neck feature fusion module, but

Table 6
Results compared to other methods.

Module	mAP _{0.5} (%)	mAP _{0.5:0.95} (%)	Params (M)	FPS	mAP _{0.5} -MAD
Faster-RCNN (Ren et al., 2016)	49.9	17.0	136.77	30	0.08
CenterNet (Zhou et al., 2019)	83.6	46.9	32.26	101	0.10
RetinaNet (Lin et al., 2017b)	76.0	45.5	36.41	53	0.10
EfficientDetv1 (Tan et al., 2020)	61.3	29.9	3.83	33	0.08
swin transformer (Liu et al., 2021)	96.8	72.0	33.09	204	0.12
RT-DETR (Zhao et al., 2024)	97.6	73.7	33.80	101	0.08
YOLOX-s (Ge et al., 2021)	92.4	54.3	58.21	106	0.08
YOLOv5-s (Jocher et al., 2020)	97.0	73.8	8.69	217	0.08
YOLOv5-m	98.1	76.0	23.89	143	0.12
YOLOv8-s	97.3	76.2	10.61	172	0.12
YOLOv8-m	97.8	78.3	24.65	125	0.10
YOLOv9-s (Wang et al., 2024c)	97.2	74.9	6.83	143	0.06
YOLOv9-m	97.6	79.4	19.22	106	0.12
YOLOv10-s (Wang et al., 2024a)	96.4	70.9	6.88	222	0.06
YOLOv10-m	98.2	76.4	14.60	139	0.08
YOLOv11-s (Khanam and Hussain, 2024)	97.0	73.5	8.98	196	0.12
YOLOv11-m	97.5	76.9	19.11	123	0.08
YOLOv12-s (Tian et al., 2025)	95.7	70.4	8.81	86	0.06
YOLOv12-m	97.6	76.0	19.18	83	0.08
LDM-PCB	99.1	81.6	8.61	100	0.08

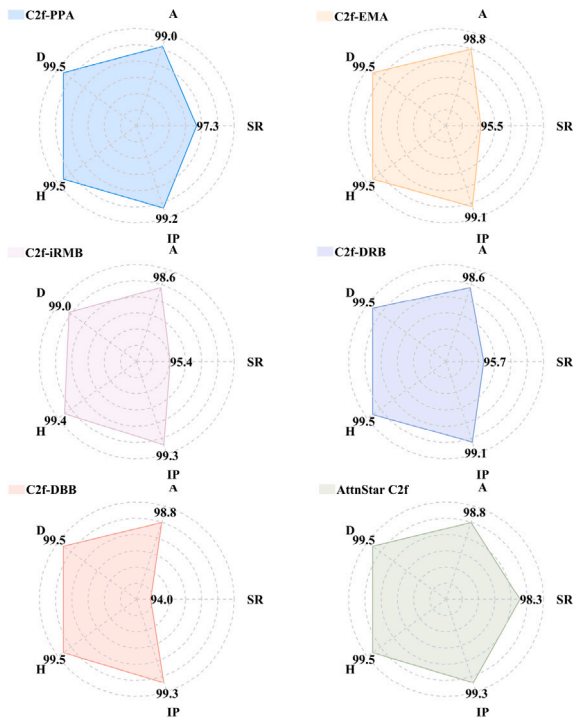


Fig. 13. Comparison of the accuracy of various defects for different C2f.

its static weight allocation mechanism limits adaptability to long-tail data, resulting in a $mAP_{0.5:0.95}$ of only 73.5%. YOLOv12s prioritizes lightweight design for edge deployment, but excessive compression compromises feature extraction capability compared to LDM-PCB, resulting in a $mAP_{0.5}$ of 95.7%. It is worth noting that even when increasing the network depth of YOLO models, their performance on both $mAP_{0.5}$ and $mAP_{0.5:0.95}$ remains comprehensively inferior to the LDM-PCB model.

The design of LDM-PCB solves the above problems: Backbone designs the LTFE-Net, which adopts element-level multiplication to construct a high-dimensional implicit feature space, combines with the attention mechanism to strengthen the ability to characterize the details of tiny defects in long-tail PCB data, and at the same time reduces feature loss by using incremental downsampling. In the Neck section, the DMS-Fuse framework is used to dynamically assign weights to achieve

Table 7
Single-category defect accuracy comparison.

Module	AP _{0.5} (%)				
	SR	A	D	H	IP
Faster-RCNN	12.8	57.2	45.5	68.9	65.1
CenterNet	49.0	91.2	84.8	97.7	95.3
RetinaNet	20.4	90.0	92.8	85.7	94.0
EfficientDetv1	9.7	71.1	78.4	57.4	89.9
swin transformer	88.8	98.2	98.9	99.5	98.5
RT-DETR	92.6	98.5	98.4	99.3	99.1
YOLOX-s	83.9	92.7	92.4	98.4	94.7
YOLOv5-s	88.5	98.5	99.5	99.5	99.2
YOLOv5-m	93.6	98.9	99.5	99.5	99.5
YOLOv8-s	89.1	98.8	99.5	99.4	99.4
YOLOv8-m	92.2	98.8	99.4	99.5	99.2
YOLOv9-s	88.9	98.6	99.5	99.5	99.3
YOLOv9-m	90.4	99.0	99.5	99.5	99.5
YOLOv10-s	88.4	97.1	98.9	99.5	98.3
YOLOv10-m	94.1	98.7	99.5	99.4	99.2
YOLOv11-s	88.6	98.2	99.4	99.5	99.3
YOLOv11-m	90.2	99.0	99.5	99.5	99.3
YOLOv12-s	83.3	97.5	99.3	99.5	99.0
YOLOv12-m	91.1	98.6	99.5	99.5	99.4
LDM-PCB	98.3	98.8	99.5	99.5	99.3

adaptive fusion of multi-scale features, which improves the ability of network to characterize 3D defect information for defects with complex geometrical patterns. The Dy-P2 layer proposed in the detection head part establishes dynamic spatial associations between different layers of dynamic multi-scale features through an adjustable sense field mechanism, which effectively alleviates the spatial-semantic disassociation problem of multi-scale features and thus improves the detection effect of tiny defects. Fig. 14 clearly demonstrates that LDM-PCB outperforms other existing methods in the trade-off between accuracy and speed, providing a reliable solution for industrial inspection.

To further evaluate the practical effectiveness of LDM-PCB in long-tail, multi-scale small object detection tasks, the single-class accuracy of various detection models across different defect categories was compared. The experimental results are shown in Table 7.

The data reveals that under long-tail distributions and multi-scale small-object scenarios, the sparse and structurally complex features of tail-end categories lead to significant variations in detection accuracy across different models for various defect classes. This is particularly pronounced for SR-class defects, where existing methods generally struggle to balance detection performance, resulting in low accuracy for this category and consequently impacting overall model performance. As a classic two-stage detector, Faster RCNN relies on predefined

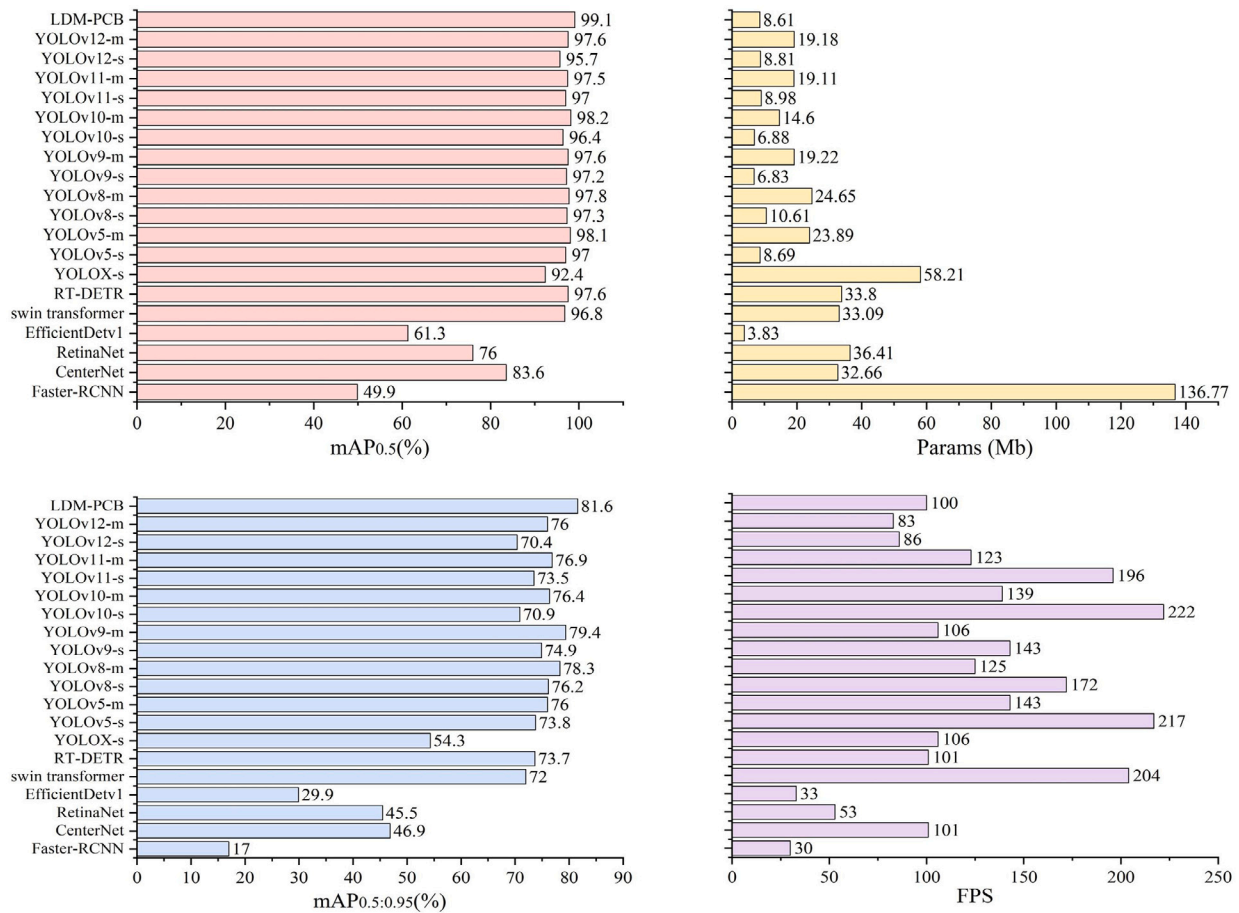


Fig. 14. Comprehensive performance comparison of various algorithms.

anchor boxes and RoI pooling operations, making it prone to losing subtle defect features and lacking long-tail optimization mechanisms. Its accuracy for SR-class defects is only 12.8%. CenterNet does not use pre-selected boxes but distinguishes objects from background through center point features. However, due to sparse tail samples leading to ambiguous feature representation, its accuracy for SR-class defects is 49.0%. RetinaNet introduced Focal Loss to mitigate class imbalance, improving SR class accuracy to 20.4%. However, it still struggles to adapt to defects with large scale variations. EfficientDetv1 employs BiFPN for efficient feature fusion but underperforms on long-tail defects, dropping SR class accuracy to 9.7%—the worst among all models. The Swin Transformer-based detection model leverages self-attention to capture global differences between defects and background, significantly boosting SR accuracy to 88.8%. However, it still trails LDM-PCB. YOLO-based models exhibit limited attention to tail defects in long-tail distributions, resulting in substantial accuracy variations across categories. While increasing network depth enhances feature extraction, it also risks losing subtle feature information. In contrast, LDM-PCB enhances feature capture for low-frequency defects through its Tail Adaptive Attention (ATA) module. By integrating adaptive weight fusion and 3D convolution techniques, it dynamically adjusts multi-scale feature weights and extracts 3D deformation information, thereby conveying rich semantic information to the detection head. This approach effectively narrows accuracy gaps between defect categories in long-tail, multi-scale small-object scenarios, mitigating the impact of long-tail distributions and scale variations on detection performance. As shown in Fig. 15, the detection performance of each algorithm across different defect categories reveals that LDM-PCB maintains relatively uniform detection accuracy even for defects with significant variations in scale across five categories. This demonstrates the model's robustness

in detecting PCB defects with long-tail, multi-scale, and small-sized objects.

4.5.2. Qualitative comparison with other methods

As shown in 16, comparing the visual inspection results of various methods reveals that LDM-PCB excels in distinguishing and identifying subtle defects with high recognition accuracy. YOLOv5 and YOLOv8 enhance computational efficiency through multiple down-sampling steps, but this reduces feature map resolution, leading to loss of fine-grained defect information and significantly increased risk of missed detections. YOLOv9 lacks targeted optimization for long-tail defects on PCBs, exhibiting insufficient recognition capability for tail-end category features. YOLOv10 employs a decoupled detection head to enhance small object detection but has limited accuracy in distinguishing feature-similar defects. YOLOv11 strengthens neck feature fusion, yet its static fusion mechanism struggles to fully adapt to long-tail distribution scenarios. YOLOv12 fails to sufficiently focus feature interactions on critical defect contours during irregular-shape defect detection, leading to misalignment between recognition results and actual shapes and causing false positives. EfficientDet-v1 achieves lightweight processing through channel pruning and resolution compression, but this significantly reduces feature extraction capability, resulting in frequent missed detections. CenterNet relies on heatmaps to locate defect centers, which are susceptible to interference in complex PCB background textures. Heatmaps generate false peaks, causing false positives. RT-DETR employs a hybrid encoder to accelerate multi-scale feature processing but remains vulnerable to noise and similar object interference in complex backgrounds, similarly facing false positive issues. In contrast, LDM-PCB introduces a Long-Tail Feature Extraction Network (LTFE-Net) to enhance perception of tail defects. Its Dynamic

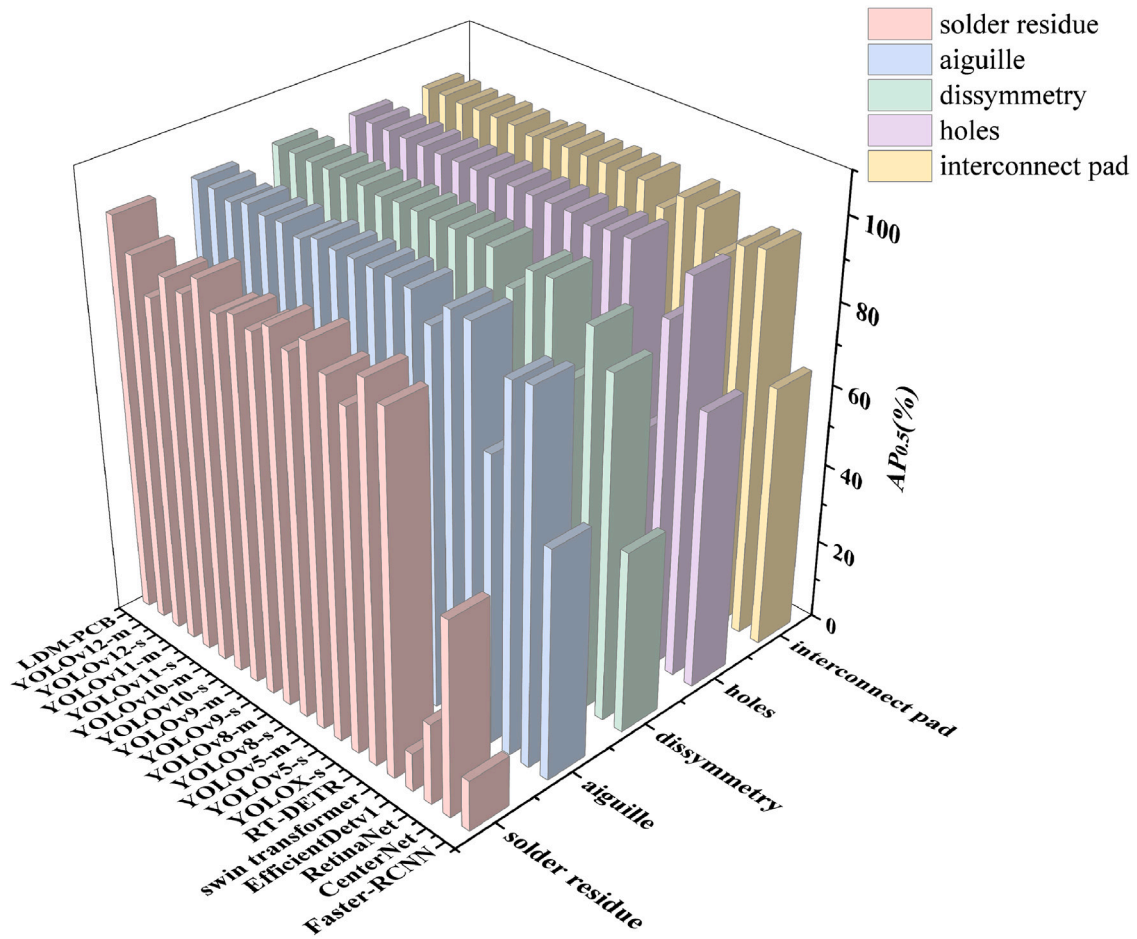


Fig. 15. Performance comparison of various algorithms for defect detection in different categories.

Multi-Scale Fusion (DMS-Fuse) architecture enables dynamic feature extraction for large, medium, and small-scale defects. By incorporating 3D convolutions to preserve depth-information-rich defect features and employing a Dynamic Prediction Layer (Dy-P2), it effectively mitigates small-object feature loss during downsampling. Visualization results demonstrate LDM-PCB's advantages in high detection accuracy and precise localization.

4.6. Comparison of heatmap based visualizations

In the field of PCB defect inspection, heatmaps are utilized as an effective visualization tool to intuitively demonstrate the degree of attention paid by the inspection model to the defect area and the intensity of the features of the defect itself. Heatmaps represent attention intensity with color gradients, with values ranging from [0,1]: red indicates higher attention or significant defect characteristics, while blue reflects lower attention or less pronounced defect characteristics. The experimental results are shown in Fig. 17, where the first row shows the real labeled GT information, which marks the actual defect location on the original PCB, as a benchmark for assessing the accuracy of the model detection. The thermogram generated by the YOLOv5 model shows that there is some thermal response in the defective area, but some of the response is more scattered and less intense. Compared with YOLOv5, the thermogram of the YOLOv8 model shows improved response in the defective area, while the response is still not concentrated enough and not strong enough for some small defects. The YOLOv5 and YOLOv8 models are able to identify the location of defects roughly, yet there are deficiencies in detail capture, and

there is still some room for improvement in accurately identifying small defects. In contrast, the thermograms generated by our proposed model show a more concentrated and strong warm response at the defect locations, indicating that our model has higher accuracy and sensitivity in the location and degree of response of the defect regions. The results show that the LDM-PCB model exhibits superior performance in the PCB defect detection task, and is capable of identifying and localizing defects on printed circuit boards more accurately.

4.7. Public dataset validation

To systematically evaluate the performance of the LDM-PCB model, this study employs the enhanced PCB defect public dataset provided by Peking University's Robot Open Laboratory. Qualitative visual comparisons and quantitative assessments of the model are conducted under identical evaluation criteria. This dataset comprises 10,668 annotated images covering six defect categories: missing holes, mouse bites, open circuits, short circuits, spurious copper, and stray copper. Specific examples are shown in Fig. 18, where missing_hole, spurious_copper, spur, mouse_bite, open_circuit, and short correspond to annotations (a) through (f), respectively.

4.7.1. Quantitative comparison with other methods

Before the experiment began, the dataset was randomly split into a training set (8640 images), a validation set (961 images), and a test set (1067 images) at a ratio of 8:1:1. To ensure fairness in the comparative experiments, all algorithms participating in the comparison were retrained using identical parameter settings. The input image size

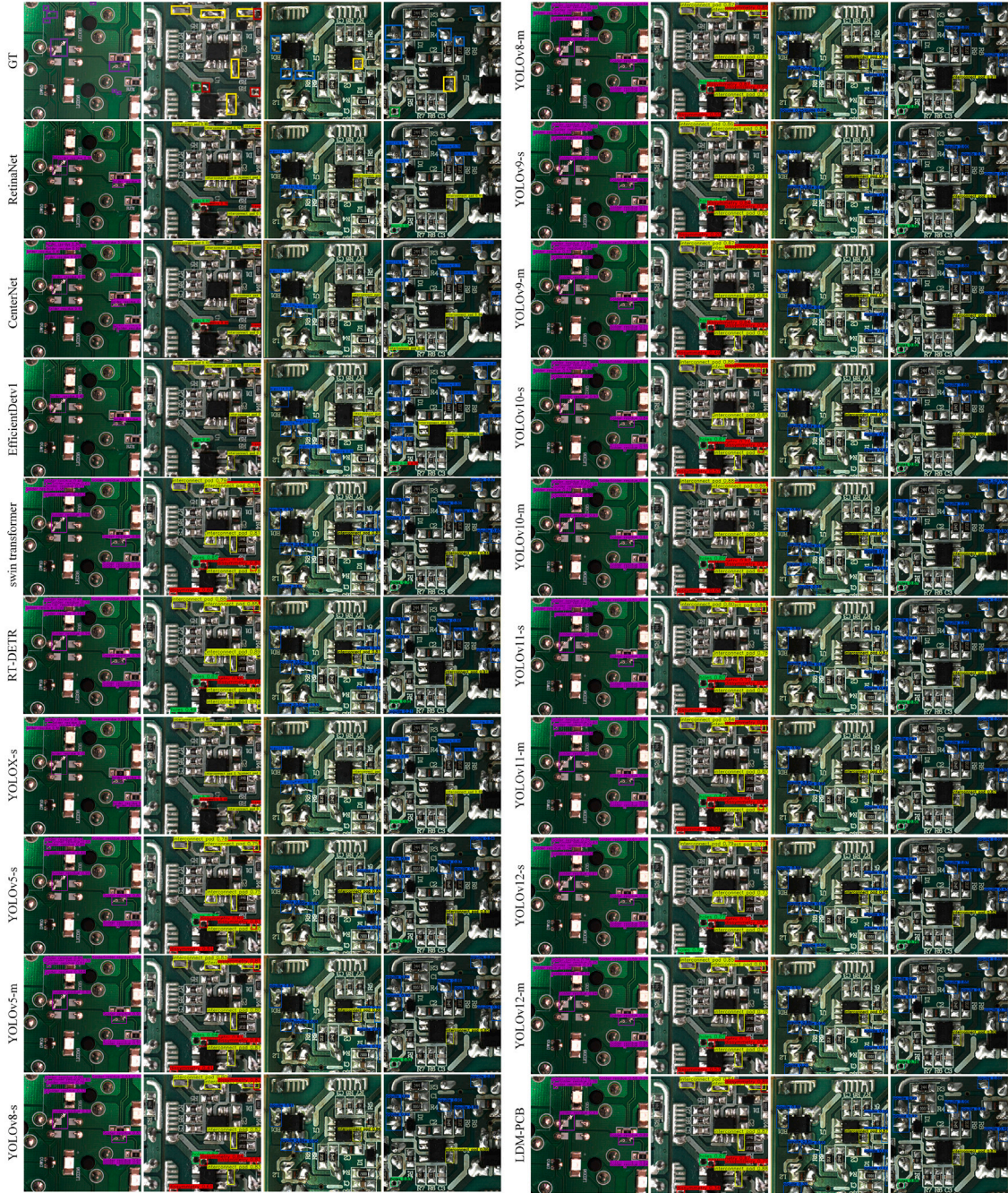


Fig. 16. Visualization of detection results for each algorithm.

was maintained at 640×640 , while other hyperparameters remained consistent with those in Table 3. The quantitative comparison results for each detection algorithm are presented in Table 8.

This experiment underwent 300 training epochs on a large-scale dataset, resulting in models with comparable performance on the $mAP_{0.5}$ metric and minimal differences. Under the more stringent evaluation metric $mAP_{0.5:0.95}$, LDM-PCB achieved the best result of 89.4%, outperforming all comparison models and demonstrating excellent comprehensive detection capability and generalization performance. Regarding detection accuracy, CenterNet avoids the anchor mechanism, but its detection mechanism based on keypoint heatmap estimation is less sensitive to minute defects, resulting in an overall $mAP_{0.5}$ of only 97.7%. Swin Transformer can model global context through self-attention mechanisms, but it inadequately captures subtle local features in PCB defects. Combined with detail loss from multiple downsampling

operations, its $mAP_{0.5:0.95}$ drops to 77.9%. RT-DETR achieves end-to-end detection via its Transformer decoder. However, its lack of long-tail distribution optimization and high parameter count (40 M) limit its industrial applicability. The YOLO series employs a unified single-stage detection framework. Its s-version models offer advantages in speed and parameter efficiency but often sacrifice accuracy for speed, with $mAP_{0.5:0.95}$ significantly underperforming LDM-PCB. The m-version model enhances defect feature perception through increased depth while also amplifying model complexity. Regarding model stability, the $mAP_{0.5:0.95}$ -MAD metric reflects performance fluctuation—lower values indicate more stable detection results. LDM-PCB achieves a MAD value of 0.08, comparable to models like Faster R-CNN, EfficientDetv1, RT-DETR, and YOLOv5s, while outperforming CenterNet, RetinaNet, and YOLOv11s. This demonstrates its robust detection capabilities. Regarding model efficiency, LDM-PCB's parameter count is

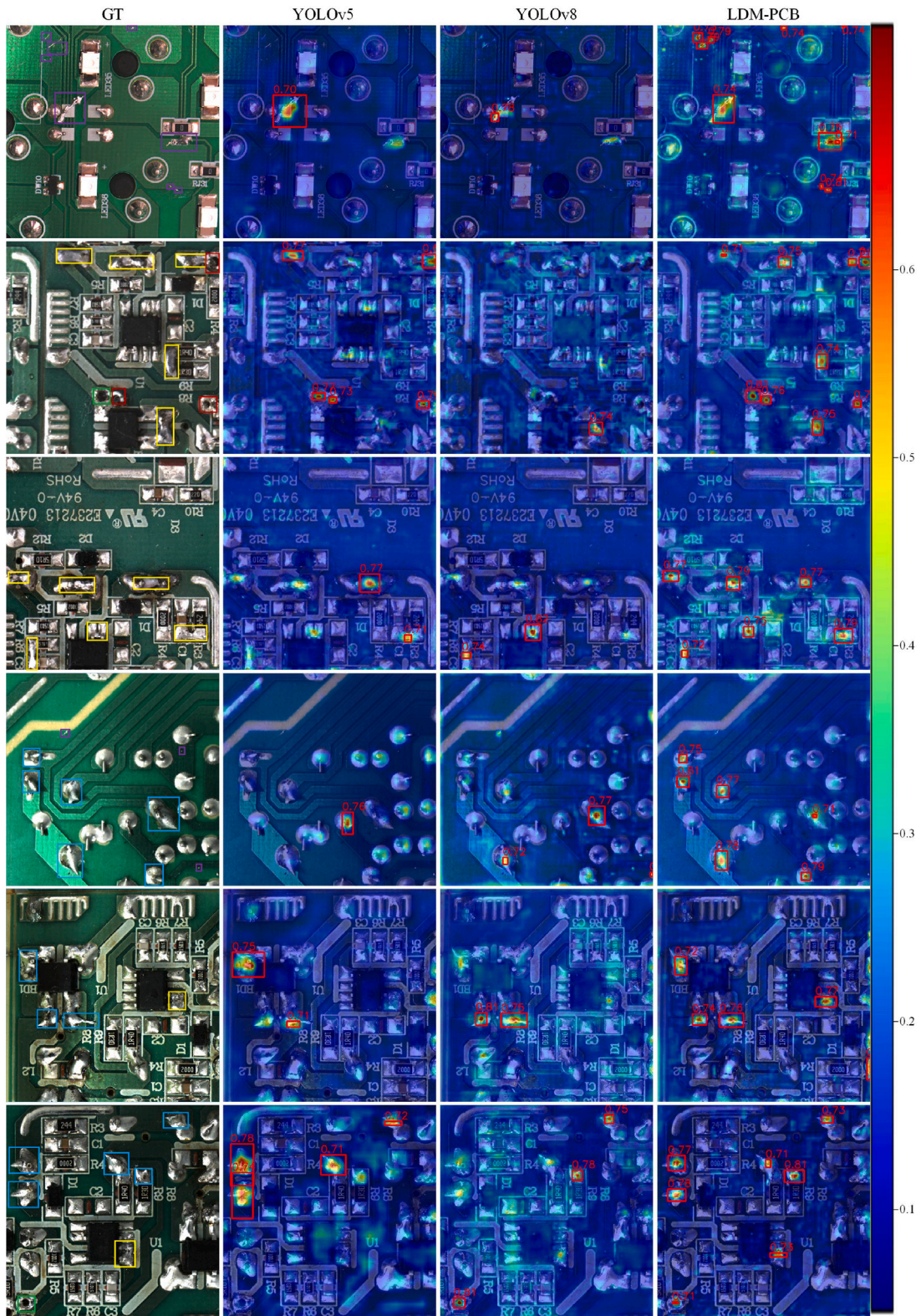


Fig. 17. Heatmap results visualization.

only 8.61 M, comparable to lightweight models like YOLOv9s and YOLOv12s, yet significantly lower than large architectures such as Faster R-CNN and RT-DETR. Although its inference speed of 116 FPS is lower than YOLOv5s and YOLOv10s, it still meets industrial real-time detection requirements, achieving a good balance between accuracy and efficiency. In summary, LDM-PCB enhances representation

and stability for rare defects through its Long-Tail Feature Extraction Network (LTFE-Net), boosts multi-scale defect detection via its Dynamic Multi-Scale Fusion Framework (DMS-Fuse), and preserves fine defect details using its Dynamic Detection Head (Dy-P2). Ultimately, it achieves high-precision detection while maintaining low parameter counts, demonstrating strong potential for industrial deployment.

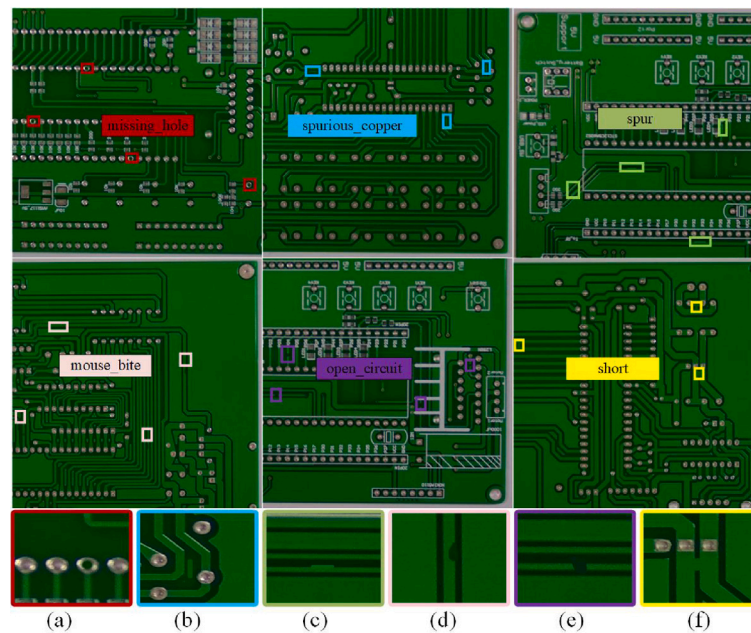


Fig. 18. Schematic diagram of PCB surface defects.

Table 8

Results compared to other methods.

Model	mAP _{0.5} (%)	mAP _{0.5:0.95} (%)	Parameters (M)	FPS	mAP _{0.5:0.95} -MAD
Faster-RCNN	72.6	28.4	136.79	30	0.08
CenterNet	97.7	52.1	32.67	101	0.12
RetinaNet	92.7	49.3	36.39	53	0.10
EfficientDetv1	85.6	43.4	3.83	33	0.08
swin transformer	99.3	77.9	33.11	75	0.12
RT-DETR	99.2	64.4	40.00	52	0.08
YOLOX-s	98.3	56.3	54.15	106	0.12
YOLOv5-s	99.3	80.7	8.69	208	0.08
YOLOv5-m	99.4	85.2	23.89	164	0.12
YOLOv8-s	99.5	85.0	10.61	142	0.10
YOLOv8-m	99.4	87.8	24.65	91	0.12
YOLOv9-s	99.2	77.7	6.89	250	0.04
YOLOv10-s	99.5	88.4	19.23	122	0.12
YOLOv10-m	99.4	83.4	14.60	78	0.08
YOLOv11-s	99.4	82.1	8.98	227	0.12
YOLOv11-m	99.5	88.3	19.11	159	0.06
YOLOv12-s	99.4	80.6	8.66	108	0.06
YOLOv12-m	99.4	84.3	18.67	104	0.08
LDM-PCB	99.5	89.4	8.61	116	0.08

4.7.2. Qualitative comparison with other methods

As shown in the visualization results of various methods in Fig. 19, LDM-PCB not only demonstrates outstanding detection accuracy in identifying common printed defects on PCB surfaces but also exhibits stable and reliable defect localization capabilities. In industrial inspection scenarios, detection accuracy and defect localization precision directly determine the quality control effectiveness of entire PCB batches, representing core requirements for industrial quality inspection. Regarding detection performance, mainstream algorithms still exhibit shortcomings: Faster-RCNN struggles with insufficient filtering and correction capabilities when background regions share similar features with foreground objects, leading to false positives; EfficientDetv1 enhances cross-scale feature correlation through bidirectional feature fusion, but becomes overly sensitive with large training datasets due to ambiguous key defect feature definitions, resulting in frequent false positives; RT-DETR possesses global feature modeling capabilities, yet its Transformer architecture exhibits limited sensitivity to subtle local features, resulting in inadequate detection performance in areas with

dense small objects. YOLOv5s achieves approximate object localization but suffers from generally low detection confidence, reflecting weaknesses in feature representation and classification stability. YOLOv11s and YOLOv11m introduce innovative network architectures, yet their learning of defect feature discrimination boundaries lacks precision. Detection accuracy remains improvable under similar texture interference and complex background conditions. LDM-PCB achieves stable defect type recognition through: - Long-Tail Feature Extraction Network (LTFE-Net) for enhanced learning of low-frequency and tail-end defect features; - Dynamic Multi-Scale Fusion (DMS-Fuse) for optimized cross-scale feature perception; - Dynamic Prediction (Dy-P2) layer for precise semantic information transfer. Evaluating performance across multiple dimensions—including detection accuracy, practical detection effectiveness, model parameter count, and inference speed—LDM-PCB demonstrates significant advantages over comparable methods. It better meets the demands for detection accuracy and reliability in industrial quality inspection scenarios, offering greater practical application value.

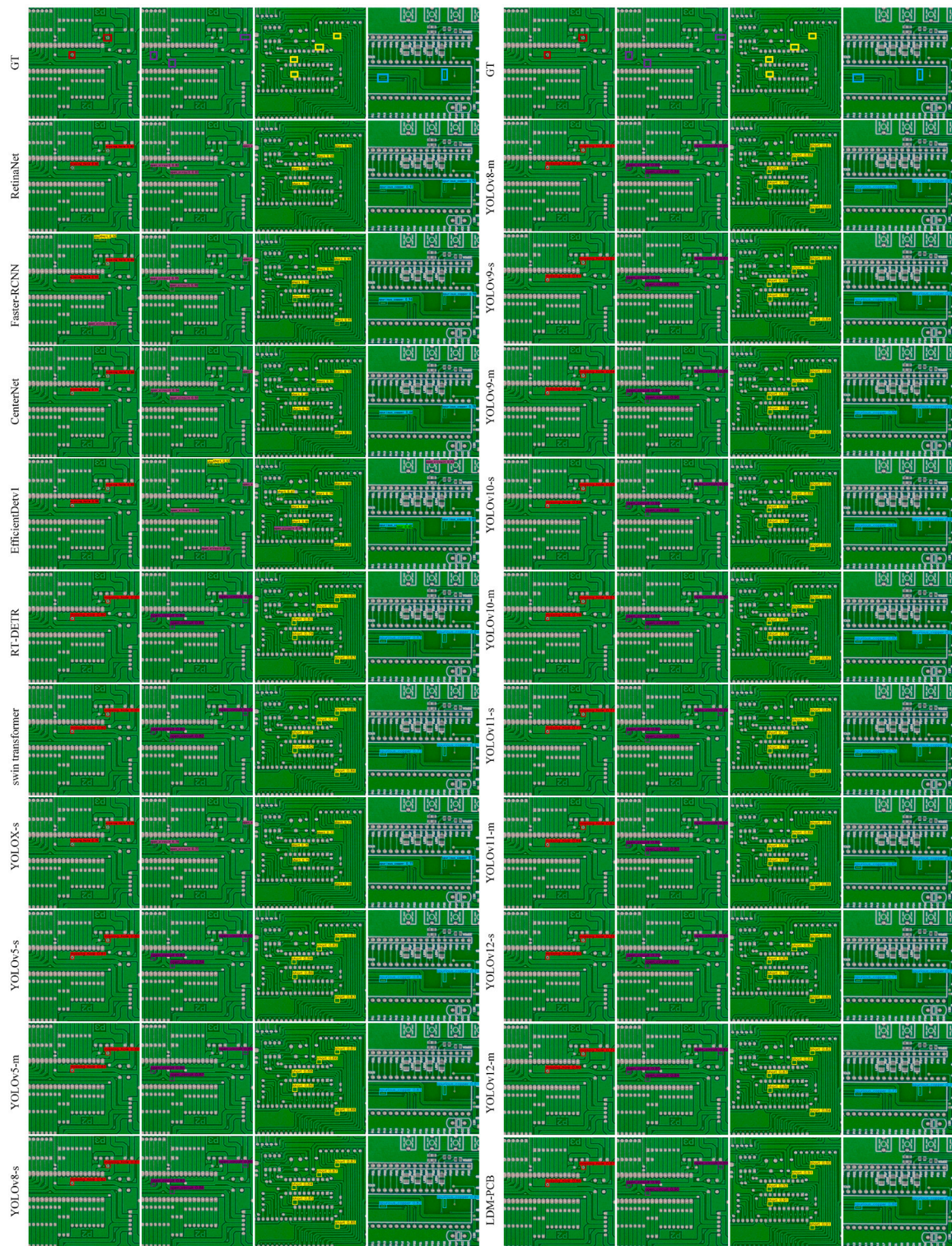


Fig. 19. Visualization of detection results for each algorithm.

4.8. Edge deployment testing

As shown in Fig. 20, to validate the deployment potential of the proposed algorithm in industrial edge computing scenarios, this study selected the NVIDIA Jetson Orin NX—whose computational power is significantly lower than that of the NVIDIA RTX 3090 Ti—as the edge hardware platform for practical testing.

This device is equipped with 16 GB of system memory, utilizes the Ampere architecture, and incorporates 1024 GPU CUDA cores. Its maximum thermal design power (TDP) ranges between 10 W and 25 W. Detailed hardware and software specifications are provided in Table 9.

Deploy the trained LDM-PCB model, comparison model, and test images onto the device. Test the model's accuracy and real-time detection performance using identical hyperparameters and environments. The

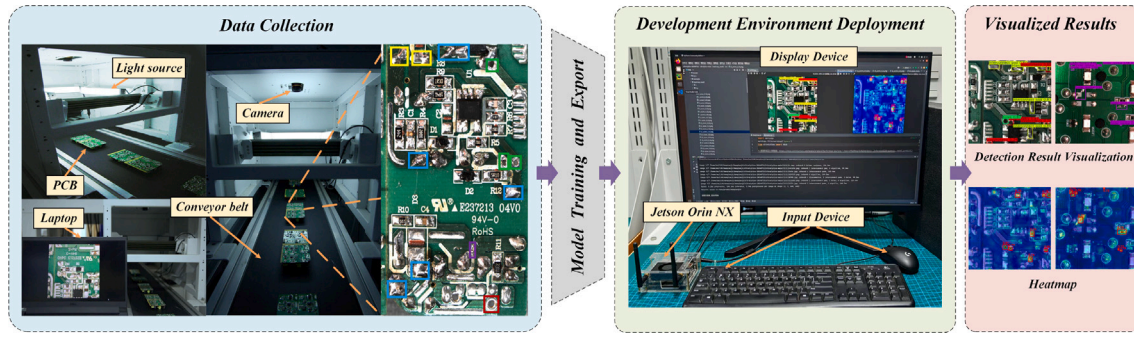


Fig. 20. Edge device deployment for each algorithm.

Table 9
NVIDIA Jetson Orin NX device specifications.

Hardware/Software category	Specific information
Device model	NVIDIA Jetson Orin NX developer kit
System memory	16 GB (GPU video memory is shared with system memory)
GPU architecture	Ampere architecture
GPU CUDA cores	1024
Maximum power consumption (TDP)	10 W–25 W
CUDA version	11.4.315
Jetpack version	5.1.1 (based on L4T 35.3.1)
Supporting library versions	cuDNN 8.6.0, TensorRT 5.1.1

evaluation metrics include data preprocessing time, model inference time, result post-processing time, FPS, and accuracy. Detailed results are presented in Table 10.

The experimental results show that the time spent on preprocessing input images is largely consistent across models, indicating that the standardized input data format and preprocessing workflow during testing ensured fair comparisons. Model inference and post-processing times reflect differences in internal structural design, which are the two primary factors affecting the FPS metric. Faster R-CNN's approach of first generating candidate boxes before performing classification regression increases model inference complexity, resulting in suboptimal detection accuracy and FPS performance that fails to meet PCB inspection requirements. CenterNet and YOLOX bypass predefined anchors to directly predict object centers, accelerating both inference and post-processing. YOLOX achieves the highest FPS at 27, but suffers from lower detection accuracy. RT-DETR and swin transformer enhance detection accuracy through global attention mechanisms, yet their FPS drops to 5, limiting real-time performance. Anchor density increases computational load in RetinaNet detection, prolonging inference time and reducing FPS to 9; EfficientDetv1 employs separable convolutions to accelerate detection but limits small-object feature extraction, lowering $mAP_{0.5}$ to 61.3%; The CSPDarknet backbone in YOLOv5-s reduces redundant computations, significantly boosting real-time detection capability with FPS reaching 23; YOLOv8-s increases branch counts, raising $mAP_{0.5}$ to 97.3%, but at the cost of slower inference; The ELAN module increases computational time during feature aggregation in YOLOv9-s; channel compression in the backbone accelerates detection in YOLOv10-s but compromises accuracy; YOLOv11-s improves real-time detection through optimized convolutions, achieving 18 FPS; YOLOv12-s requires recalculating anchor offsets, substantially increasing computational load and slowing post-processing. As YOLO models increase in depth, detection accuracy improves but real-time performance diminishes. LDM-PCB achieves the highest accuracy among all compared models with a $mAP_{0.5}$ of 99.1%, while maintaining 10 FPS. This strikes an excellent balance between detection accuracy and inference speed, making it suitable for edge hardware environments with low computational power and low power consumption. It demonstrates strong potential for deployment in medium-to-low-speed industrial edge detection scenarios.

5. Conclusion

This paper addresses challenges in industrial PCB surface defect detection, including the long-tail distribution of randomly occurring defects, the difficulty of fixed multi-scale feature fusion strategies in adapting to size-varying defects, and the loss of fine-grained features due to continuous downsampling. It proposes LDM-PCB, an improved long-tail dynamic multi-scale detection model based on YOLOv8. The backbone network introduces LTFE-Net, which constructs high-dimensional defect features through an Adaptive Tail Attention (ATA) module. This enhances focus on tail defects, mitigating training bias caused by class imbalance in long-tail distributions. The neck module employs the DMS-Fuse architecture, dynamically adjusting feature fusion weights for large, medium, and small defects via adaptive weighting strategies. It integrates 3D convolutions to extract defect depth information, improving multi-scale defect representation. The head module's Dy-P2 layer employs a high-resolution retention mechanism. This directly maps multi-scale fused features to the output while minimizing detail loss, thereby improving localization accuracy for minute defects. Experiments on a self-built long-tail PCB dataset demonstrate that LDM-PCB achieves a favorable balance between detection accuracy, model compactness, and inference speed. Compared to the baseline model YOLOv8-s, it improves $mAP_{0.5}$ by 1.8% to 99.1%, increases $mAP_{0.5:0.95}$ by 5.4% to 81.6%. Parameter size is reduced by 18.6% to 8.61 M, while inference speed remains stable at 100 FPS. For long-tail data, multi-scale defects, and small object detection tasks, LDM-PCB demonstrates optimal overall performance compared to 19 other mainstream algorithms. Simultaneously, LDM-PCB exhibits outstanding generalization and robustness on the PKU-Market-PCB augmented dataset, with potential for deployment on edge devices. Despite its strong experimental performance, LDM-PCB has limitations requiring future refinement. First, training and validation were based solely on five typical soldering defect categories, excluding common issues like corrosion, component misalignment, and cold soldering. This limits its generalization capability in real production lines. Second, defect samples are generally scarce in natural settings. Limited training data may hinder the model's stable performance across diverse defect morphologies, lighting conditions, and background interferences, thereby constraining its generalization in practical detection scenarios. Additionally, existing research primarily focuses on two-dimensional image defect detection, with only preliminary exploration of three-dimensional defect features incorporating height and depth information, failing to fully leverage three-dimensional defect data. To address these challenges, future work will focus on three key areas:

(1) Building Diverse Datasets: Future efforts will construct more comprehensive industrial defect datasets by incorporating multi-type defect samples and integrating 3D visual sensors to capture high-precision depth information. This will establish multimodal architecture datasets to enhance defect detection robustness.

(2) Generative Data Augmentation: Introduce generative AI-based unsupervised defect detection frameworks to augment sparse datasets,

Table 10
Comparison of reasoning times for various algorithms.

Name	Pre time (ms)	Inf time (ms)	Post time (ms)	Total (ms)	FPS	mAP _{0.5} (%)
Faster-RCNN	5.2	327.5	9.8	342.5	3	49.9
CenterNet	5.4	68.2	4.7	78.3	13	83.6
RetinaNet	5.3	103.0	5.7	114.0	9	76.0
EfficientDetv1	5.3	90.6	5.3	101.2	10	61.3
swin transformer	5.0	191.2	4.9	201.1	5	96.8
RT-DETR	5.2	202.5	2.4	210.1	5	97.6
YOLOX-s	5.2	27.0	4.4	36.6	27	92.4
YOLOv5-s	5.4	34.9	3.8	44.1	23	97.0
YOLOv5-m	5.4	56.4	3.4	65.2	15	98.2
YOLOv8-s	5.7	37.6	3.5	46.8	21	97.3
YOLOv8-m	5.4	73.7	5.6	84.7	12	97.8
YOLOv9-s	5.3	44.0	3.2	52.5	19	97.2
YOLOv9-m	5.4	88.6	4.6	98.6	10	97.6
YOLOv10-s	5.2	32.6	1.7	39.5	25	96.4
YOLOv10-m	5.3	61.9	1.8	69.0	14	98.2
YOLOv11-s	5.4	46.2	4.6	56.2	18	97.0
YOLOv11-m	5.2	74.2	5.2	84.6	12	97.5
YOLOv12-s	5.3	51.4	5.2	61.9	16	95.7
YOLOv12-m	5.4	104.1	4.2	113.7	9	97.6
LDM-PCB	5.3	92.0	3.3	100.6	10	99.1

mitigating model training and generalization challenges under conditions of scarce defect samples.

(3) Integration of 3D Vision and Multimodal Data: Integrate 3D vision sensors to capture high-precision depth information and construct multimodal datasets, aiming to achieve more accurate and robust defect detection.

CRedit authorship contribution statement

Xuangang Li: Writing – original draft, Validation, Software, Methodology. **Sen Wang:** Methodology, Funding acquisition, Formal analysis. **Liyang Zhu:** Software, Data curation. **Aiping Shen:** Validation, Software. **Dianlu Hu:** Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.engappai.2025.113415>.

Data availability

Data will be made available on request.

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